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COMPUTER MAINTENANCE AND TECHNICAL SUPPORT MODULE

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14 February 2023

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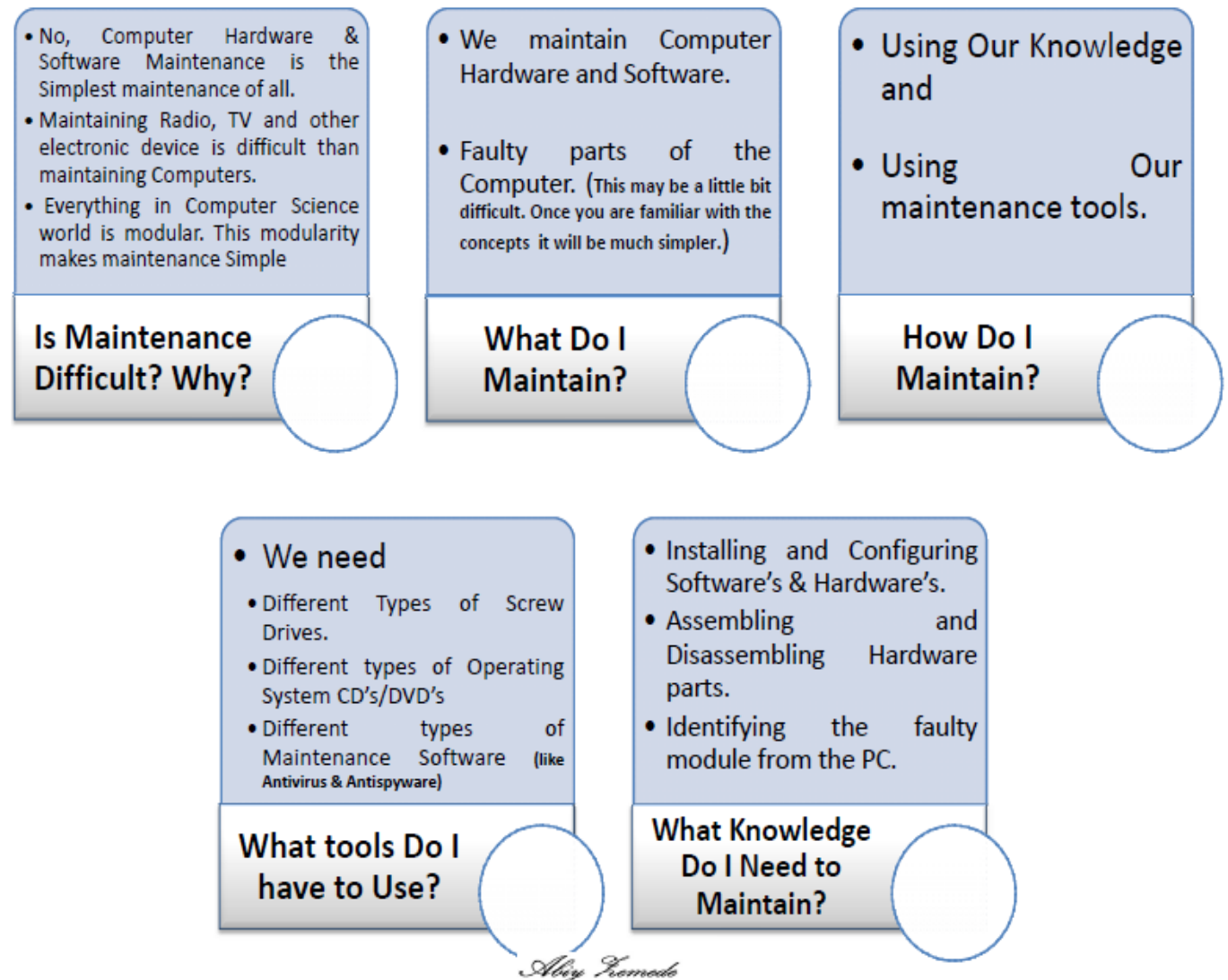
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Chapter 1: Maintenance Basics

1.1. General concepts about PC and Technical support

WHAT IS MAINTENANCE?



Effective troubleshooting of electronic equipment is a matter of combining good knowledge of the equipment and its operation with good testing techniques and deductive reasoning skills. In general, the process of troubleshooting microprocessor-based equipment begins at the outside of the system and moves inward. The first step is always to try the system to see what symptoms are produced. Second, you must isolate the problems to either hardware or software related problems. After this, the problem should be isolated to a section of the hardware or software. Finally, the problem must be isolated to the offending component.

1.2. Tools and Workspace

Anyone who is going to work on any type of equipment must have the proper tools for the task.

Tool	Comments
Standard screwdrivers (not magnetic)	High-quality Phillips screwdrivers are indispensable. At the very least, you'll need small and larger Phillips screwdrivers with various shaft lengths. A few different sized flat-head screwdrivers are very helpful to have. Do not use magnetic screwdrivers inside computers.
Cable testers	Testers are available for most types of cables used with a computer. While there are other ways to test cables, such as swapping them with known good cables, cable testers save time and trouble.
Anti-static (ESD) wristbands	Tool to protect computer circuits against the damage even carpet shocks can cause.
Anti-static (ESD) mats	Provides a static-free surface. Can be used with anti-static wristbands. Includes anti-static floor mats.
Internet access	Allows access to Web-based virus-scanning software and other utilities and easy downloads of device drivers.
Data transfer cables	Cables of various types such as serial, parallel and USB that allow for different methods of data transfer.
Cleaning and maintenance tools: vacuums and dust-cleaning sprays	Computers get dusty inside and dust build-up interferes with proper cooling. Use sprays such as Blow Off while vacuuming to clean out the dust.
Uninterruptible Power Supply (UPS)	These provide continuous power to a computer when there is a power failure.
Data-recovery software	Software that can often recover data from damaged hard drives.
CD/DVD kits	It is used to clean CD-ROM drive

1.2.1. General Advice

There are some basic recommendations that will make repairs run as smoothly as possible and ensure good post-repair performance

How to Spot Quality Components

How do you determine which parts are high quality? It's not necessarily a matter of selecting the most expensive components. There are many clues to help guide you in this matter. The most obvious are that the components come with some type of manual or guide and the company supplies you with a way of contacting them for support or warranty needs. The manual might be in the form of a file on the installation CD and the contact might be an address, a telephone number, or more likely these days just a Web site or e-mail address. You want to be able to reach customer service and tech support. Some vendors have excellent customer service and tech support.

Be Careful with Components

Hardware components are susceptible to physical and electrical damage due to mishandling and power surges. Follow these guidelines to minimize the risk of damage:

Make sure the power supply is set to the correct voltage: In Ethiopia, electric utilities provide 220–240 volts AC to wall outlets. Check the power supply at the back of the computer. Most have switches labeled 120/240. Make sure the voltage setting matches the actual voltage, or you could face catastrophic results.

Use surge suppressors and UPSs whenever possible: Power surges and under voltages can damage or even destroy components. **Surge suppressors** is electrical device inserted in a power line to protect equipment from sudden fluctuations in current

Don't force components: Connectors should be pushed in directly, not at an angle, and most go in only one way. Match up the pins with the holes. If you try to force a connector in the wrong way, you can bend or push in pins, damaging the connector and most likely causing the device to malfunction. In the case of a device with a permanently connected cable, which is common with monitors, damaging the connector requires that a qualified monitor technician repair the monitor.

There are different procedures for inserting expansion cards, memory chips, and CPUs, although the same "no-force" warning applies.

Prevent static damage: Discharge as much static as possible by touching a ground—any large metal object. Use anti-static floor mats, surface mats, and wrist ground straps. Make sure all circuits are wired correctly, including grounding.

Document all changes: Keep a record of every change you make; you might have to undo certain changes. Mark all jumper and wire positions, before changing them, with a fine-tipped permanent marker. Make notes and diagrams of wire and jumper positions and keep a record of all software configuration changes

Safety Measures and Principles

- For any work you do on a computer, you must have an adequate workspace. The work area must be flat. If it's not, small parts could roll around and possibly get lost or damaged.
- Shut down any running programs and turn the computer off. Remove all cables (especially the power cable) that are attached to the computer.
- Wear an ESP wrist Strap before opening any System Unit.

Disassembly Prerequisites

1. **Shut down** any running programs and **turn the computer off**.
2. **Remove all cables (especially the power cable)** that are attached to the computer. Remember that some cables use special screws to attach them to their ports. If you disassemble the computer while it's plugged in or turned on, you could get electrocuted! Additionally, components could be damaged if inserted or removed while power is applied.
3. **Remove any floppy disks** from their drives to **prevent damage** to either the disk or the drive.
4. After checking once more to see that all the prerequisites have been dealt with, move the computer to the work surface.

Removing the Case Cover

1. This is accomplished by **removing some (but usually not all) of the screws** at the back of the computer and **slowly pulling the computer's case cover back and up to remove it**. Different casing has different ways they can be removed. Most casing has screws. The Dell OptiPlex for example comes with a **hub-handle** with two (big) buttons at both ends (pressed down) to release hook on the side of the casing.
2. If you **pull too fast, it is possible to damage cables** that may be located close to the top of the case.
3. If the **case doesn't move**, check to see that all the case cover **screws were removed**.
4. **Don't just start removing all the screws at the back of the computer!**

Some of these screws hold vital components (such as the power supply) to the case and removing them will cause those components to drop into the computer.

The computer's documentation should indicate which screws to remove in order to remove the case cover.

Removing the Expansion Cards

Once you have the case off, you will be able to see the inside of the computer. The next step in disassembly is to **put on an antistatic wrist strap**, plugging one end into the ground plug of an outlet. Then you can start to remove any expansion cards, mainly because they are the easiest things to remove at this point.

1. Remove any internal or external cables or connectors. (You should already have removed the external ones in the prerequisite steps.) When you are removing a cable or connector, use your pen and notepad to diagram their installed positions. Also, use the pen to make identifying marks on the cables to make sure they align correctly when reinstalled.

2. Remove any mounting screws that are holding the boards in place and place the screws somewhere where they won't be lost. If you don't have an egg carton you can put each screw back into its mounting hole as you remove each expansion board. Just make sure to screw them in all the way, not just a couple of turns.

3. Grasp the board by the **top edge with both hands and gently rock it front to back** (NOT side to side). If the expansion board doesn't come out easily, don't force it. You may damage it.

4. Finally, once the board is out, **place it in an antistatic bag to help prevent ESD damage** while the board is out of the computer. Place the board aside and out of the way before continuing. Repeat the procedure for each card.

Removing the Power Supply

The next component that should be removed is the power supply. You can easily access it with the expansion cards out of the way. Before you remove the power supply from the computer, however, you must do two things: **disconnect the power supply connectors from the internal devices**, and **remove the mounting hardware for the power supply**.

1. Disconnecting the connectors from the devices is a very easy process. Before removing any power connectors, **note their positions and connections** on your notepad (and make any marks on them to indicate their installed positions). Then, **simply grasp the connector (NOT THE WIRES)** and gently wiggle it out of its receptacle. **Then, proceed to the next connector. The system board and disk drives both use power connectors.**

2. The standard PC power supply has **two connectors to the motherboard**. These plug into receptacles that are side by side. Document the positioning of these connectors before removing them to make it easier to put them back when you are done. If, for some reason, you get confused as to which connector goes where, the general rule is **black-to-black**. Grab the connectors so that the black wires are side by side, and you will be holding them the way they should be installed.

3. Once all the **power supply connectors are disconnected from their devices**, you can **remove the mounting hardware for the power supply**. In most PCs, you can detach the power supply from the case by **removing four screws**. Some power supplies **don't need to have screws removed; instead, they are installed on tracks or into slots in the case and need only to be slid out or lifted out**.

Removing the Disk Drives

Most disk drives are installed in IBM-compatible computers with **rails that are attached to the drives with screws**. These rails **allow the drive to be slid into the computer's drive bays like a drawer**. The drives are then secured with at least two screws on the sides.

1. Many desktop computers use a **special drive carrier** that holds the drive in place and can be **easily removed without tools**, thus eliminating the need for special rails.
2. With most drives, however, you can just remove the mounting screws and slide the drive out. Consult the computer's documentation to see exactly how to remove your specific type of drive.
3. To remove drives from higher-end computers, you should also consult the documentation. **It is possible on many higher-end machines to swap higher drives while the computer is still running**; to prevent damage to the drive (or maybe even other components), you will want to know the correct procedure.

Removing the Motherboard

All that's left in your computer is the motherboard (also known as the **logic board** or **Planner Board** or **System Board**).

1. The only time the motherboard should need to be **removed is when it needs to be upgraded or replaced**. Otherwise, you should leave the motherboard in the PC's case to **prevent either physical or ESD damage**.
2. The motherboard is held away from the metal case using **plastic spacers and is secured and grounded using mounting screws**.
3. To remove the motherboard, **you must remove the screws holding the motherboard to the mounting brackets**. Then, you must slide the motherboard to the side to release the spacers from their mounting holes in the case.

1.3. Static energy and its effect on computers

Static electricity is **the imbalance of electric charge on a surface of a material**. Static means fixed or stationary, therefore it is used in contrast to dynamic (moving) electricity which are in the form of electric currents. Typically, atoms are neutral, which means they have the same number of electrons and protons.

1.4. Preventive maintenance and troubleshooting

Preventive maintenance can be the key to keeping computer systems from experiencing serious problems, such as data loss and hardware failures, and it also helps systems have a longer life span.

In this section, you study the need for preventive maintenance of a computer system. Following a good preventive maintenance plan can keep computer problems from being too troublesome.

Preventive maintenance is a regular and systematic inspection, cleaning, and replacement of worn parts, materials, and systems. Preventive maintenance helps to prevent failure of parts, materials, and systems by ensuring that they are in good working order.

Troubleshooting is a systematic approach to locating the cause of a fault in a computer system. A good preventive maintenance program helps minimize failures. With fewer failures,

there is less troubleshooting to do, thus saving an organization time and money. Preventive maintenance can also include upgrading certain hardware or software such as a hard drive that is making noise, upgrading memory that is insufficient, or installing software updates for security or reliability.

Troubleshooting is a learned skill. Not all troubleshooting processes are the same, and technicians tend to refine their troubleshooting skills based on knowledge and personal experience. Use the guidelines in this chapter as a starting point to help develop your troubleshooting skills.

Explain the Purpose of Preventive Maintenance

Preventive maintenance reduces the probability of hardware or software problems by systematically and periodically checking hardware and software to ensure proper operation.

Hardware

Check the condition of cables, components, and peripherals. Clean components to reduce the likelihood of overheating. Repair or replace any components that show signs of damage or excessive wear.

Use the following tasks as a guide to create a hardware maintenance program:

-  Remove dust from fan intakes.
-  Remove dust from the power supply.
-  Remove dust from components inside the computer.
-  Clean the mouse and keyboard.
-  Check and secure loose cables.

Software

Verify that installed software is current. Follow the policies of the organization when installing security updates, operating system updates, and program updates. Many organizations do not allow updates until extensive testing has been completed. This testing is done to confirm that the update will not cause problems with the operating system and software.

Use the tasks listed as a guide to create a software maintenance schedule that fits the needs of your computer equipment:

-  Review security updates.
-  Review software updates.
-  Review driver updates.
-  Update virus definition files.
-  Scan for viruses and spyware.
-  Remove unwanted programs
-  Scan hard drives for errors.
-  Defragment hard drives.

Benefits

Be proactive in computer equipment maintenance and data protection. By performing regular maintenance routines, you can reduce potential hardware and software problems.

Regular maintenance routines reduce computer downtime and repair costs.

A preventive maintenance plan is developed based on the needs of the equipment. A computer exposed to a dusty environment, such as a construction site, needs more attention than equipment in an office environment. High-traffic networks, such as a school network, might require additional scanning and removal of malicious software or unwanted files. Document the routine maintenance tasks that must be performed on the computer equipment and the frequency of each task. This list of tasks can then be used to create a maintenance program.

The following are the benefits of preventive maintenance:

-  Increases data protection
-  Extends the life of the components

- ✚ Increases equipment stability
- ✚ Reduces repair costs
- ✚ Reduces the number of equipment failures

Chapter 2: Computer Cases and the Motherboard

2.1. Computer cases

A computer's case is more than a box. The case includes the drive cage. An assembly in a computer that holds disk drives, the internal compartment that holds disk drives, and almost always the power supply, among other features. Cases come in various types called form factors, which differ in layout of components. The case's form factor needs to match that of the motherboard and the power supply. Just as with motherboards, some cases are proprietary and require proprietary power supplies and motherboards.

Case Components

Cases come with various components



Figure 1. A typical case

Most of components of case are self-explanatory. Drive bays are areas in the front for installation of removable media (CD, DVD, floppy, Zip, etc.). Cases usually come with small speakers, which are there to provide very basic sounds to the user. About all these speakers play are warning beeps and the sound of a modem connecting. These speakers are very important, because multimedia speakers (Sound cards) don't work before Windows has booted, if Windows is in Safe Mode or if there is a problem with the sound card.

I/O Shields

I/O shields are the metal plates that surround the ports that are built into the motherboard. They are used to shield (protective covering or structure) these ports from radio frequency interference (RFI) and to provide openings in the case in the correct size and configuration for the motherboard's built-in ports.

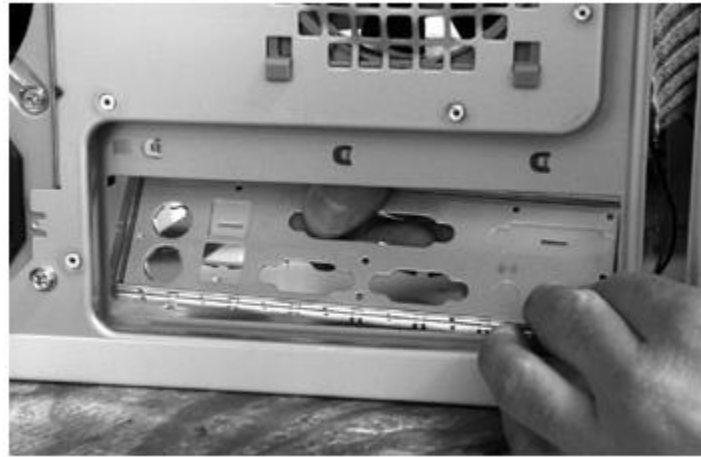


Figure 2. Attaching an I/O shield

Case Quality

A good quality case can have a big impact on the performance and durability of the computer. Cases sold have to meet requirements for shielding against both external RFI and interference to external devices that is generated by the computer. Good cases are designed to provide for proper airflow to keep the components from overheating. Panels should fit together well without requiring excessive effort. When the computer is running, a good case does not make excessive vibration noises. Other attributes of good cases include ease in opening and ease in accessing internal components.

Case Form Factors

Cases come in many form factors, the most common being AT and ATX.

Opening Different Types of Cases

Cases also vary in the difficulty of opening and accessing internal components. While most cases open easily after removing screws from the back. Some manuals tell you how to open the case.

One common style of case requires you to remove the left-side panel only as shown the following figure.



Figure 3. Removing a side panel

Note: Different Casing has different ways they can be removed. Most casing has screws at the back. For example, Dell OptiPlex comes with a hub-handle with two (big) buttons at both ends (pressed down) to release hook on the side of casing.

2.2. System board defined

The motherboard is the part of the computer to which every other component is connected. It contains the processor socket(s), memory slots, expansion card slots, ports for mouse, keyboard, printer, etc. and electronic parts, known as the chipset, to make everything run. Most motherboards contain some built-in components such as video, sound, network adapter and others.

- ✚ A motherboard is a multi-layered printed circuit board. It is sometimes called as Planner Board/ System Board.
- ✚ Think of a motherboard as a scale model of a futuristic city with many modular plug-in buildings
- ✚ The motherboard is the data and power infrastructure for the entire computer.
- ✚ The form factor is essentially the size, shape and design of the actual motherboard.

Form Factors

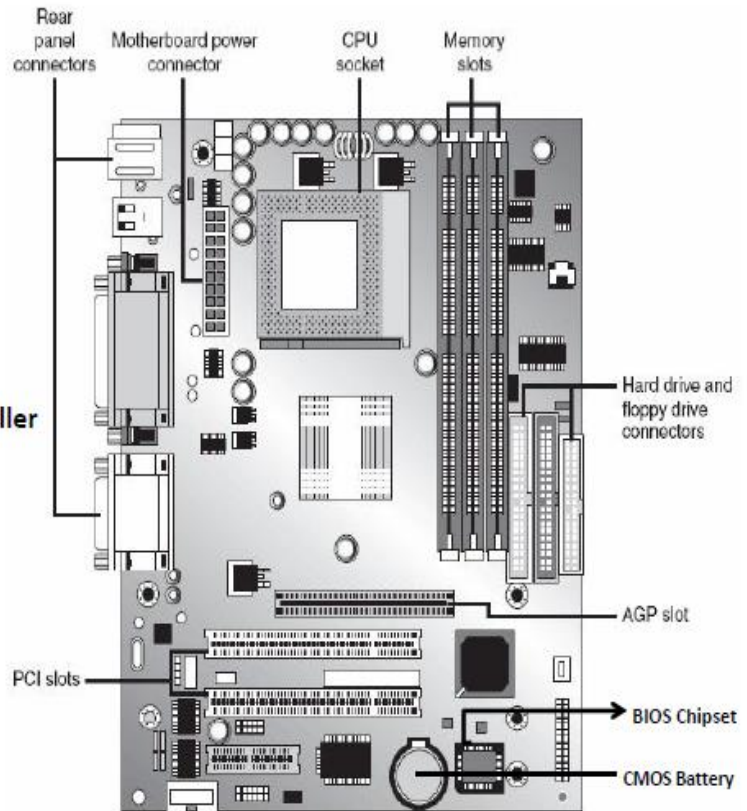
Since the first PC was introduced, several types of motherboards have been used; the types referred to as form factors. What differentiates form factors of motherboards is their size, arrangement of components on the boards and other details. Cases and power supplies are also classified into the same form factors; they all have to match to some extent for the components to fit properly into the case. The following form factors of motherboards are among those that have been used for PCs: AT and ATX.

The original computer was the IBM AT computers. The AT motherboard has specific layout which is called the form factor but the modern system now have ATX form factors. The form factor is important because the motherboard must have a **casing** that matches the form factor of your board. But the most common type of casings we have presently are ATX-casings.

The kind or type of casing you have depends on what you want on your computer. Some casings have USB ports in front of them and even presently you have some casings which have alternative keyboard port on the front. Some have only one diskette bay and some an extra bay for diskette. There are also bays for your CD drives, Hard Disk etc.

Contents of the Motherboard

- Expansion Slots
- Memory Slots/RAM Banks
- Processor/CPU socket/slot
- Chipsets
- Ports
- On Board Disk Drive Interface/Controller
- BIOS Chip
- CMOS Battery
- Power connectors
- Jumpers and DIP switches



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Figure 1: A typical motherboard

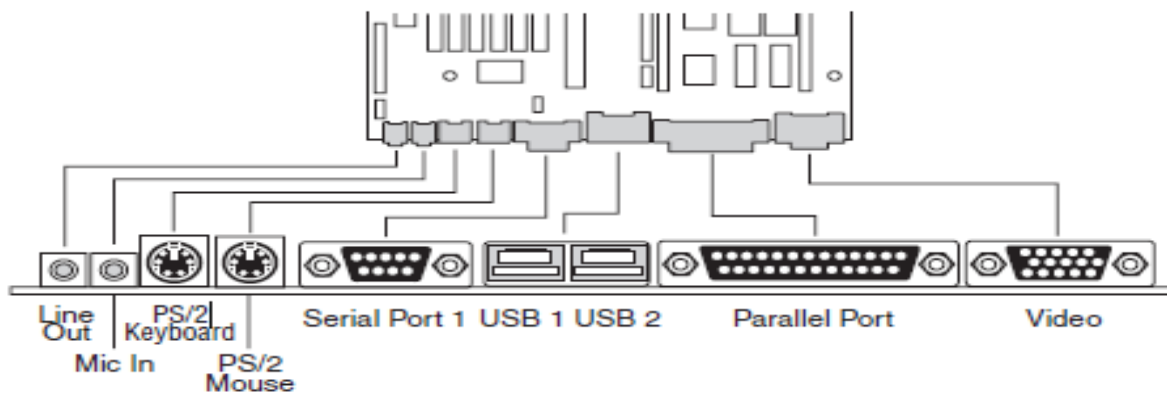


Figure 2: Motherboard back panel connector

Motherboard Components

It is important for technicians to be able to identify the parts of any motherboard.

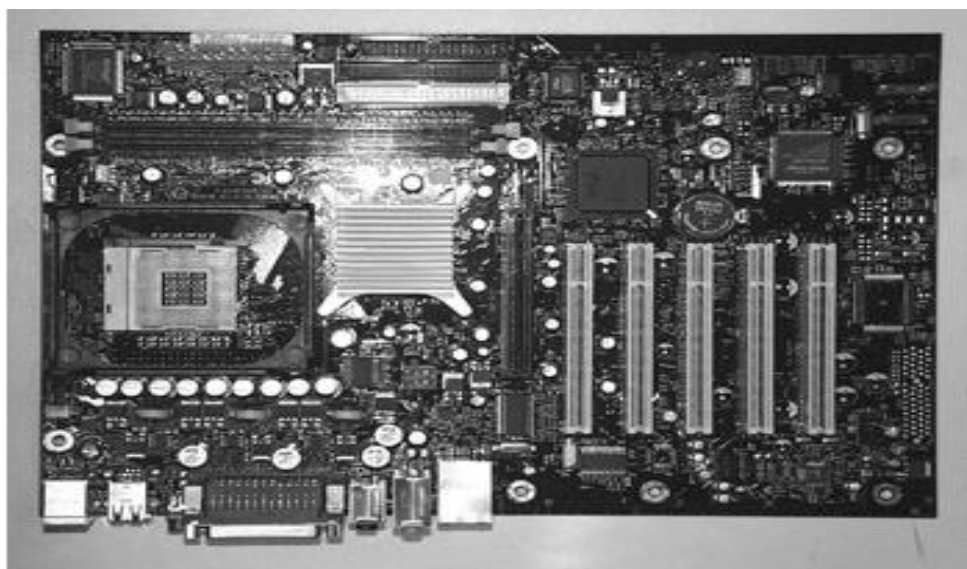


Figure 3: ATX motherboard

While there are wide differences among the various brands and form factors, all motherboards have certain components in common:

Memory slots: Physical configurations of memory chips have changed over the years, but the industry seems to have settled on dual inline memory modules (DIMMs). These chips have 72 or more pins per side, although the two sides are so close to each other that it is not readily apparent that there are two sets of pins. Memory slots can accept one type of memory chip.



Figure 4: DIMM slots

Chipset: Every motherboard has a number of integrated circuits (chips or ICs) permanently installed on different parts of the board. Each chip has a separate function. It is most common for these chips to all be from a single manufacturer.

AGP slot: All motherboards made in the last several years that don't have built-in video and some that do, have an accelerated graphics port (AGP) slot. This is the slot for a video adapter.

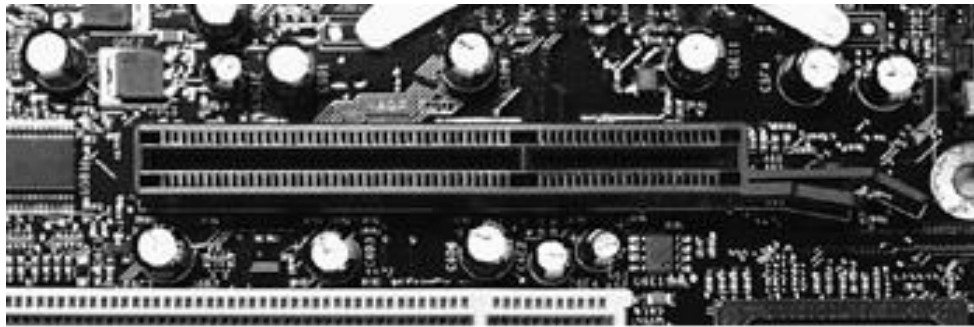


Figure 5: AGP video slot

Expansion slot (PCI and ISA): Expansion devices in card form, such as modems and network adapters, go into these slots. Newer motherboards have only Peripheral Component Interconnect ([PCI](#)) slots, while some middle-aged boards have a combination of PCI and the older Industry Standard Architecture ([ISA](#)) slots. Some boards have expansion slots that give a choice between the two, as shown in Figure 6. ISA slots are often dark colored and PCI slots are usually white or off-white. The cards are also easy to differentiate; compared with ISA pins, PCI pins are much smaller and closer together.

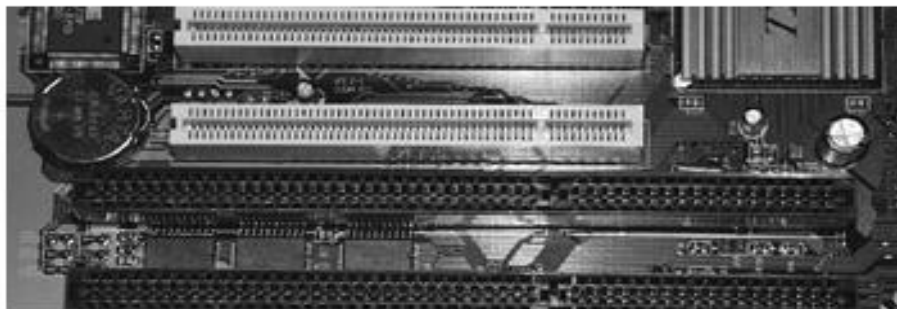


Figure 6: A combination ISA/PCI slot

Power connectors: Every motherboard has power connectors that look something like the one shown in Figure 7. Some newer boards have a 12-volt connector like the one in Figure 8. The one in Figure 8 must be connected to the proper connector on the power supply for the motherboard to receive power. If the board has a 12-volt connector, it must be connected to the power supply to avoid damage to the board.

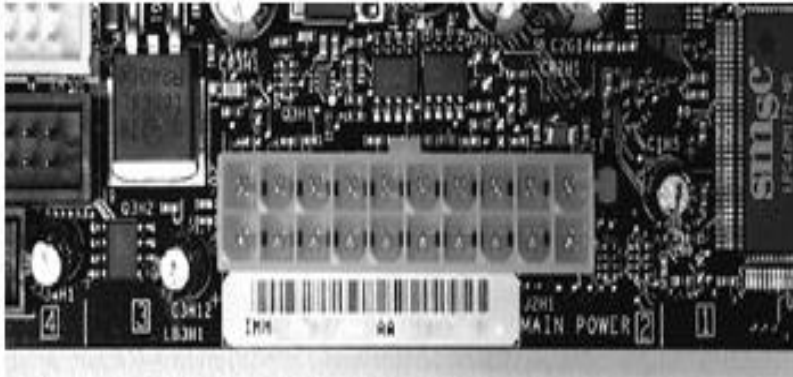


Figure 7: ATX power connector

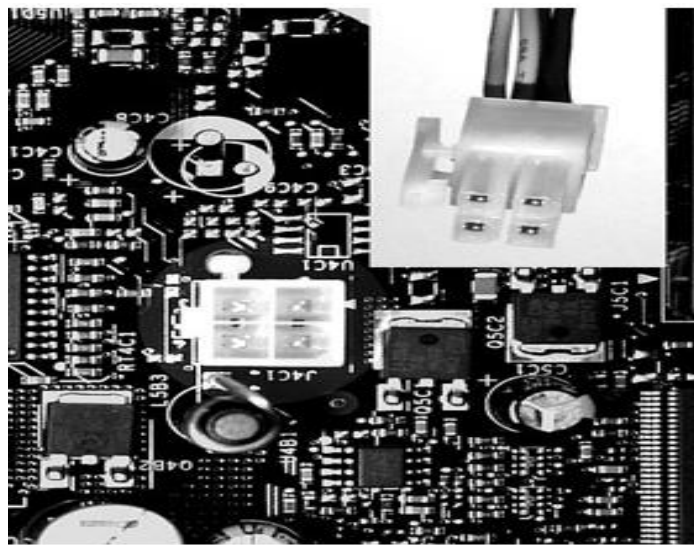


Figure 8: If the motherboard has this 12-volt connector, it must be connected. The connector from the power supply is in the inset.

An AT (form factor) motherboard power connector is different from an ATX connector. The AT connector has two parts, each with black wires on one end. They must be installed with the black wires next to each other at the center of the motherboard's connector, as shown in Figure 9.

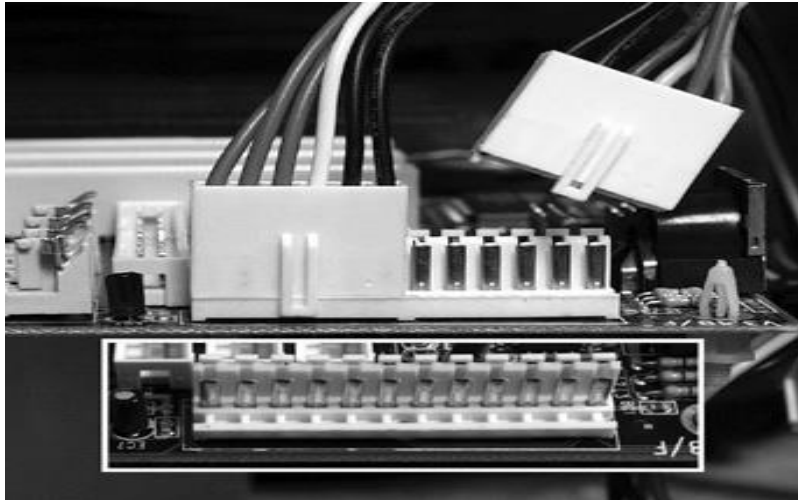


Figure 9: AT power connector

Disk drive connectors: Virtually every motherboard has two IDE connectors for up to four IDE devices, usually one or two hard drives and one or two optical (CD or DVD) drives. With the proper cables, each connector can support two drives. In addition, there is a connector for the floppy drive.

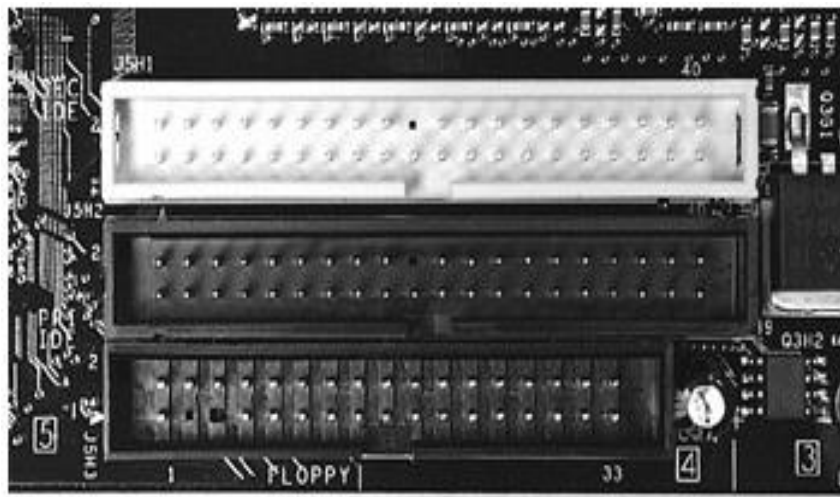


Figure 10: IDE and floppy disk drive connectors

Ports: Motherboards have some or all of the following ports: serial, parallel, PS/2 mouse, PS/2 keyboard and USB.

Identifying a Motherboard

While many motherboards are easily identifiable, a few aren't. Here are some identification methods:

- ✚ Look for the brand name, model number and revision number printed on the circuit board.
- ✚ Look for a sticker underneath the lowest expansion slot. It might not be visible without disassembling the computer and removing the board or at least by using a small mirror on a handle (preferably nonconductive).
- ✚ On bootup, look on the first screen—if the information does appear, you won't have long to see it unless you press the Scroll Lock key.
- ✚ The information might appear somewhere in the BIOS.
- ✚ If the computer is a brand-name computer, you can often find the board used by going to the computer manufacturer's Web site.

Selecting a Quality Motherboard

While the chipset manufacturer can make a difference in the quality of a motherboard, the manufacturer of the board itself makes the most difference. To select the best, most appropriate board, first decide on the features the user needs such as type and number of expansion slots plus the needed built-in components. Also note that motherboards take only one type of memory. Generally, the faster the memory, the more expensive it is. Therefore, decide on the best memory you can afford and select the motherboard accordingly.

Next, evaluate the manufacturer based on its available technical support and its Web site. Web sites should have technical specifications and information, plus driver and BIOS downloads and these should be easy to find and download, if needed. Then, consider warranty and cost. Another useful method is to go to the processor manufacturer's Web sites for recommendations of motherboards (intel.com, amd.com).

Built-In (Onboard) Components

Just about all motherboards these days have some built-in components that were available only on expansion cards on earlier PCs. The most common of these are video, sound, network adapters and modems. It is easy to tell when a board has these systems built in—just look for the appropriate connectors. If the computer is fully assembled, look at the back to determine the functions that are built in. The computer shown in Figure 11, for example, has built-in video and sound.



Figure 11: A computer with built-in sound and video.

Certain instances call for replacement of a built-in component, usually when that component fails or when the user desires a better component or one with more features. It is practically impossible to replace the individual parts on the motherboard, so the answer is to add an expansion card with the desired function. For example, if the built-in video system becomes troublesome, you can add a PCI or AGP video card. On some boards, you can add a video card and still use the built-in video in order to use two monitors.

Removal and Replacement of Motherboards

When would you need to replace a motherboard? There are a few situations; for example, when the board fails or the user wants to upgrade a good quality computer.

Note: Unless the new motherboard is identical to the old one, most or all drivers will be different. For this reason, you'll want to back up data on the hard drive containing Windows, if necessary, and format (erase) the hard drive before going any further. When you have installed a new board, be prepared to install Windows and all programs from scratch and to restore data from the backup.

When replacing a motherboard, there are some general precautions (safety measures) to take and procedures to follow

Removing the Existing Motherboard

First, with the power off, disconnect every cable from the outside of the computer. After opening the case, make sure to use a grounding strap and perhaps other anti-static devices in order to reduce electro static discharge then, disconnect all cables you can access from inside the computer, including the disk drive cables and small audio cables. You will most probably have to remove the drive cage. Remove any remaining cables and all cards in expansion (PCI, ISA, AGP, etc.) slots and place them on an anti-static surface. It is probably best to leave the processor and memory in their places for now. Now, remove the screws holding the motherboard to the case and carefully remove the board. If it is still usable and/or the CPU and memory is still in place, place it on the anti-static surface.

Installing the New Motherboard

First, make sure that the new motherboard is the same form factor as the case. Then, make absolutely certain that the power supply is set for the correct voltage to avoid zapping (strike at with power) the new board.

Next, make sure that no conductive surface comes in contact with any metal parts of the case. While some cases have elevated mounting holes that hold the board away from the case wall (see Figure 12), other cases call for standoffs. Standoffs are small spacers that go between the board's and the case's mounting holes (see Figure 13).



Figure 12: Elevated mounting holes.



Figure 13: Standoff assortment

Diagnosing Motherboard Problems

The most obvious sign of a damaged motherboard is a burnt or otherwise visibly damaged part. Naturally, motherboards can die without any visual signs. If you are sure the power supply works and is turned on, set for the correct voltage, and is connected correctly, and the CPU is good, but you attempt to boot and absolutely nothing happens, it is likely that the motherboard is dead. The only answer here is to try to replace the board with one that is compatible with the case, CPU, power supply, memory, and peripherals.

Motherboard Batteries

Motherboards come with batteries, usually replaceable lithium coin cells. These batteries are in place primarily to keep the time/date clock running and maintain BIOS setup program settings. The batteries usually last at least six years. If a computer loses its BIOS settings and time/date memory every time it is shut down, you'll need to replace the battery, if possible. The POST might show an error message if the battery has died. Moreover, if someone has set a supervisor password in the setup program, and now has forgotten it, and the computer cannot boot, the only way to reset or cancel the password is to remove and replace the battery.

2.3. Chipset

The chipset is the motherboard; therefore, any two boards with the same chipsets are functionally identical.

The chipset usually contains the processor bus interface (called front-side bus, or FSB), memory controllers, bus controllers, I/O controllers, and more.

All the circuits of the motherboard are contained within the chipset.

The chipset represents the connection between the processor and everything else. The processor can't talk to the memory, adapter boards, devices, and so on without going through the chipset.

The chipset is the main hub and central nervous system of the PC.

- ✚ If you think of the processor as the brain, the chipset is the spine and central nervous system.
- ✚ Because the chipset controls the interface or connections between the processor and everything else, the chipset ends up dictating which type of processor you have; how fast it will run; how fast the buses will run; the speed, type, and amount of memory you can use; and more.
- ✚ The chipset of a motherboard refers to the collection of semiconductor chips that provide interfaces for expansion cards, memory, peripherals, and interfaces.
- ✚ The make and model of a chipset depends on a particular Original Equipment Manufacturer (OEM). Each chipset is designed to offer certain features such as onboard audio and video functionality. Chipsets are divided into two

Northbridge

- ✚ The Northbridge is a chipset is a technique used to allow communication among high-speed peripherals such as memory, the Peripheral Component Interconnect (PCI) bus, the Accelerated Graphics Port (AGP) bus and so on.
- ✚ The Northbridge communicates with the processor using a front side bus (FSB). The actual performance of the motherboard depends on the performance of the Northbridge chipset. It also manages communication with the Southbridge.

Southbridge

- ✚ The Southbridge chipset controls all of the computer's onboard Input/Output (I/O) functions such as USB, PS/2, parallel, serial, wired, and wireless LAN ports, and IDE, audio, and so on. Southbridge usually consists of a single semiconductor chip.

2.4. Removing and replacing motherboard

2.4.1. Remove the motherboard

To remove the motherboard, first you remove the touch mat, the right and left rear covers, the column assembly, the bottom rear cover, the motherboard EMI shield, the wireless LAN board, the system memory, the thermal module, and the CPU using the following instructions:

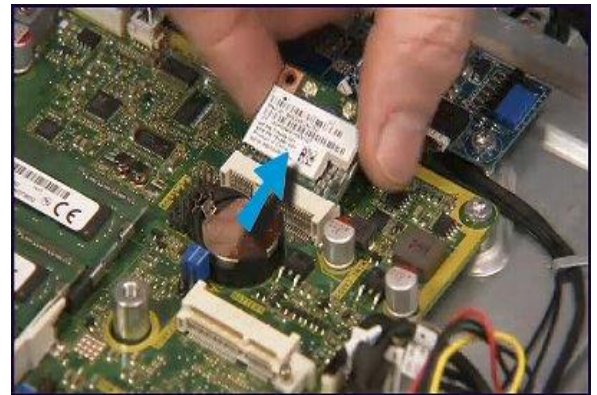
Step 1: Remove the parts to access the motherboard EMI shield and remove the motherboard EMI shield.

For detailed instructions to remove the motherboard EMI shield, see the figure.



Step 2: Remove the wireless LAN board.

For detailed instructions to remove the wireless LAN board, see the figure.



Step 3: Remove the system memory.

For detailed instructions to remove the system memory, see the figure on the right.



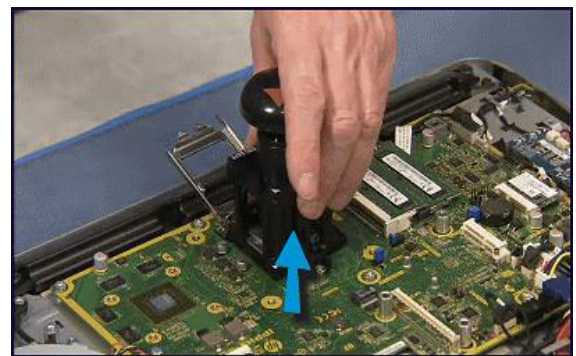
Step 4: Remove the thermal module.

For detailed instructions to remove the thermal module, see the image.

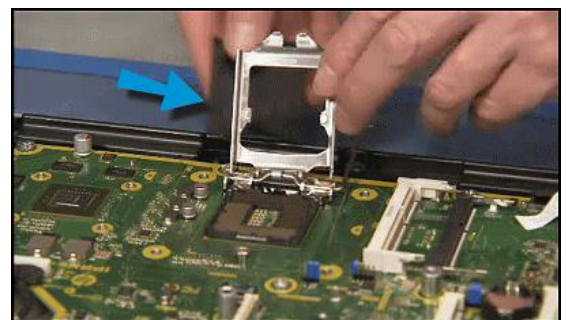


Step 5: Remove the CPU.

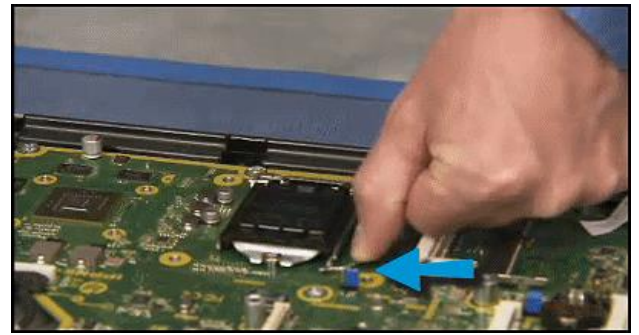
For detailed instructions to remove the CPU, see the image.



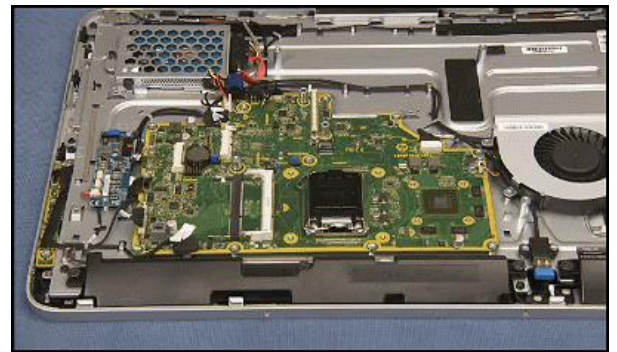
Step 6: Install a socket cover on the load plate window after removing the CPU...



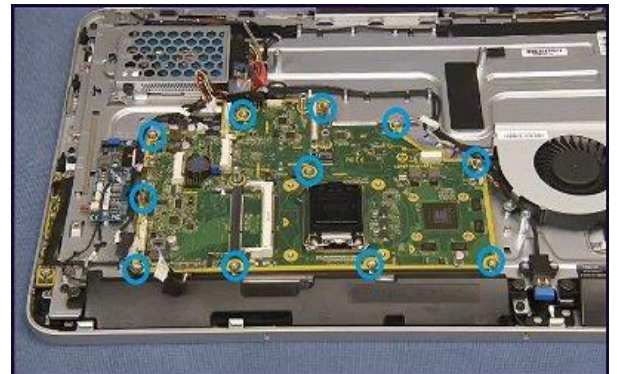
...and lower the load plate over the socket and lock it into place.



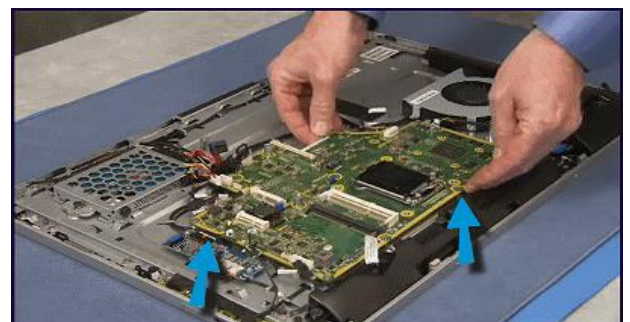
Step 7: Detach all system connectors to the motherboard.



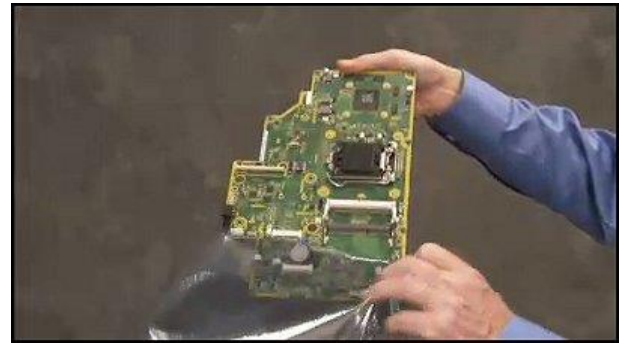
Step 8: Remove the eleven silver torx head screws securing the motherboard to the middle frame.



Step 9: Lift the motherboard off of the computer. Carefully lift with enough force to release the motherboard from the adhesive thermal pads.



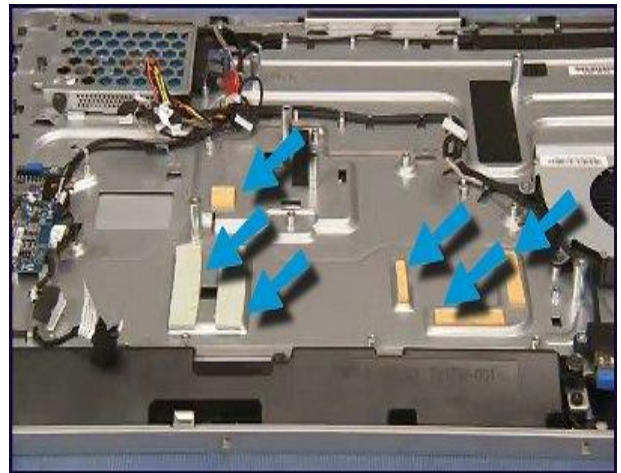
Step 10: Place the motherboard in an antistatic bag.



2.4.2. Replace the motherboard

Use the following steps to replace the motherboard:

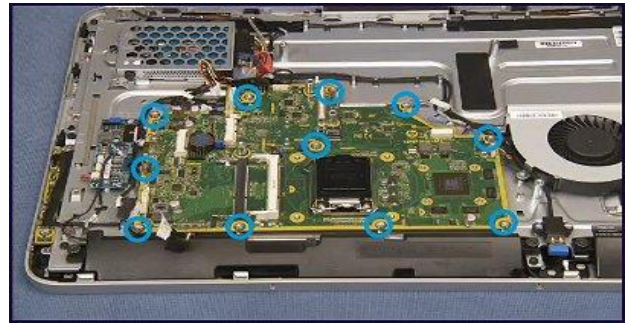
Step 1: To replace the motherboard, first make sure all thermal pads are correctly placed on the middle frame under the motherboard.



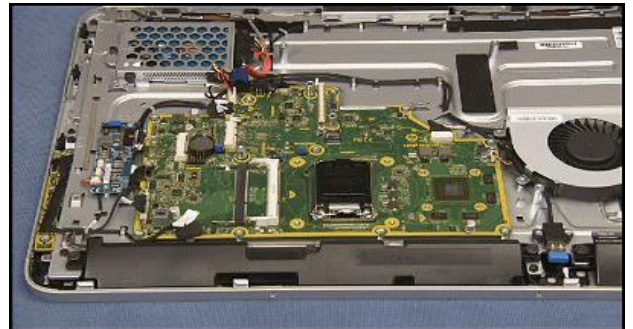
Step 2: Align the screw holes on the motherboard with the standoffs on the computer, clearing any connectors away from the side of the board.



Step 3: Replace the eleven silver torx head screws to secure the motherboard to the middle frame.



Step 4: Reattach all system connectors to the board.



Step 5: Replace the CPU.
For detailed instructions to replace the CPU, see the image.



Step 6: Replace the thermal module.
For detailed instructions to replace the thermal module, see the image.



Step 7: Replace the system memory.

For detailed instructions to replace the system memory, see the image.



Step 8: Replace the wireless LAN board.

For detailed instructions to replace the wireless LAN board, see the image.



Step 9: Replace the parts that were removed to gain access to the motherboard EMI shield.

For detailed instructions to replace the motherboard EMI shield and the rest of the parts, see the next figure.



2.5. Troubleshooting the motherboard

The motherboard is the heart and brains of your computer. It is also a complicated and delicate piece of hardware that impacts everything else in your computer. Motherboard issues can be difficult to troubleshoot, especially for those who are newer to electronics and PC building. Many components are located on the motherboard, so it can be tricky to discern which part is causing the error.

It is possible to address some of the more common motherboard issues at home however. Here are four common motherboard repair problems and solutions everyone can try:

1. Check the power

If your computer doesn't turn on or if it turns off randomly, you may be experiencing power issues. The well-worn advice of "checking the power" is especially true for motherboards because they require not one, but two power connections. Just because something is plugged in and you see a green power light turn on doesn't mean that your motherboard is receiving adequate power. Ensure that both the 20-to-24 pin connector and the four-or-eight pin connector hidden away by the processor socket are properly seated (plugged in) and matched with the proper supply cords. Unplugging them and plugging them back in is also worth a try.

2. Overheating

Overheating is a common problem that will cause your computer to run slowly, freeze, or crash frequently. In the worst cases, an overheated motherboard or other related components will have to be replaced entirely.

Start off by powering off the computer and allowing it to cool. Turn off demanding applications, which will reduce the computing power required to run the device. Placing the computer on a hard and flat surface with lots of space around it will also promote airflow, as opposed to placing the computer on carpet or inside a cabinet.

If the problem persists, try using a blower to clean the computer of dust and dirt, which can trap heat inside your computer. For optimal performance, perform internal cleaning every six months to promote airflow. Attaching an external cooling system to your computer case in order to blow cold air inside can also help prevent your PC from overheating.

3. Incorrectly installed components

The motherboard is connected to many other components, such as the video card, processor, and RAM. If improperly installed, they can cause issues. Maybe the computer boots up but there is no picture or video. In some cases, your computer may even fail to power on.

You can give the video card a push down to ensure it's secure in its slot. Check that there are no bent pins in the card or the socket. Again, check that all the power cables are securely connected to both the video card and the motherboard. Your RAM sticks should be sitting securely in their slots with the wings fastened securely over each side. Processors are a bit more delicate. Check that yours is lined up correctly with the motherboard.

4. A short circuit

Motherboards are filled with capacitors (electrical components that store energy) and connections that route data and power from one place to another. If you look at your computer, you'll see that your motherboard is installed in a way where it is held up and away from the PC case. This prevents the motherboard from touching other metal components, which could cause a short circuit that would damage the capacitors. A motherboard with a short circuit may have trouble booting up, or may crash randomly. Check your motherboard to ensure that it is installed in a way that prevents it from contacting the case. Capacitor repair however, may be best left to experts since it requires soldering.

These are some of the more common and straightforward [motherboard repair](#) issues. Learning the basics for how to address and repair motherboard issues can empower you to build and repair your own PC, and maintain optimal performance over a longer period of time. In many cases however, it's best to leave motherboard repairs to the experts. At Case model, we have expert technicians on hand who can accurately diagnose and repair motherboard and other computer issues. If your computer is having issues, visit us at one of our locations across BC and Alberta and we'd be happy to give you a free consultation.

2.5.1. Problems during Upgrading a Motherboard and Troubleshooting Techniques

Physically installing the motherboard and its connected components is a relatively straightforward process, but booting the system after that installation is not.

Because the motherboard is essentially the backbone of any PC, you can track many system errors to either the motherboard itself or a connected component. But tracking those errors can be extremely tricky. Luckily, there are telling signs for many errors and this article will dissect them to get you on the path to error-free computing.

As with any system troubleshooting process that involves physically opening your computer case, always make sure you shut down your system, unplug your computer and all of its components, and ground yourself before venturing into a computer's internal space. We recommend grounding yourself by wearing an antistatic wristband or touching a metal object outside your system's case to discharge any static electricity.

Problem: Your new motherboard doesn't fit in your computer case.

Solution: A component that is often overlooked (not taken into account) in the upgrade process is the case itself. If your new motherboard does not fit in your case, there is not much you can do other than buy a new case.

Problem: The computer won't boot at all

Solution: Make sure that the system cable from the power supply is inserted properly into the motherboard.

Problem: The power cable appears to be properly inserted, but the computer still won't boot.

Solution: One of usual suspects when troubleshooting motherboard upgrades is a loose cable. Double-check all cables, including ribbon cables, ensuring that each inserts correctly into the motherboard and its connected devices. Better yet, detach each cable one at a time and reinsert them. Just because a cable appears securely fastened to a slot doesn't mean it is. Many times a cable that is too short or isn't otherwise given enough slack (cable that is hanging loosely) will pull itself from a slot, so be sure there is enough room within the computer case and sufficient cable length to attach devices properly.

Poor attachments aren't the only possible cable problem. When upgrading your motherboard, hopefully you followed the basic rules of keeping your computer case neat. If you didn't, disorderly cables might interfere with crucial system devices, such as the processor fan. In general,

make sure that cables (and their end attachments) are not in contact with the processor fan, power supply fan and any fans for peripherals.

Problem: All of the cables are attached properly, but the computer still won't boot.

Solution: When you installed your motherboard, did you over-tighten the screws used to attach it to the case? Check for damage around the screws; a screw that is attached too tightly can cause internal damage to the board's circuitry. Also, check to make sure that the screws aren't touching any wires or cables from the motherboard.

Problem: The cables and screws are fine, but the computer still won't boot. What else could be wrong?

Solution: The memory you used with your last motherboard isn't necessarily compatible with your new motherboard, even if the modules fit properly within the slots. Consult your motherboard's manual to ensure you are using the proper memory modules. If your memory type is correct, try removing the modules from the motherboard and reinstalling them.

Problem: The memory is installed correctly, but the computer still won't boot.

Solution: A poorly seated processor will always spell trouble during a motherboard upgrade. If you are unsure whether your processor (including its sink and fan) is properly inserted, remove it from the motherboard. If you are placing a processor into a socket configuration, make sure the bottom of the processor is clean and no pins are bent. After aligning the pins with the socket's pinholes, set the processor into place and gently push. Push the socket levers into place and plug the power supply into the motherboard.

Problem: The processor is installed correctly, but the computer still won't boot. Now what?

Solution: Although modern motherboards are made to withstand a decent amount of pressure, they are still delicate devices that should be treated with care. If you notice a crack in your motherboard (perhaps from inserting components too forcefully), you might have a serious problem on your hands. If the motherboard is not operating (as in, the system isn't booting) and you notice that the board is cracked, this is most likely the cause. Unfortunately, your only option in this situation is to replace the motherboard.

Chapter 3: Central Processing Unit (CPU)

A microprocessor or a central processing unit (CPU) is a specific chip that controls your entire computer system. The microprocessor is the brain inside your system. It processes all the instructions necessary for your computer to perform its duties. The more powerful the microprocessor chip, the faster and more efficiently your system runs.

Microprocessors carry out the various instructions that let your computer compute. Every input and output device hooked up (connect or link) to a computer the keyboard, printer, monitor, and so on either issues or receives instructions that the microprocessor then processes.

Processor Specifications

Processors can be identified by two main parameters: how wide they are and how fast they are. The speed of a processor is a fairly simple concept. Speed is counted in megahertz (MHz) and gigahertz (GHz), which means millions and billions, respectively, of cycles per second and faster is better! Three main specifications in a processor are expressed in width. They are

-  Data (I/O) bus
-  Address bus
-  Internal registers

Note that the processor data bus is also called the front side bus (FSB), processor side bus (PSB), or just CPU bus. All these terms refer to the bus that is between the CPU and the main chipset component. Intel uses the FSB or PSB terminology, whereas AMD uses only FSB.

Data I/O Bus

Perhaps the most important features of a processor are the speed and width of its external data bus. This defines the rate at which data can be moved into or out of the processor.

The processor is the external data bus the bundle of wires (or pins) used to send and receive data. The more signals that can be sent at the same time, the more data can be transmitted in a specified interval and, therefore, the faster (and wider) the bus. A wider data bus is like having a highway with more lanes, which enables greater throughput.

Data in a computer is sent as digital information in which certain voltages or voltage transitions occurring within specific time intervals are used to represent data as 1s and 0s. The more wires you have, the more individual bits you can send in the same time interval. All modern processors from the original Pentium through the latest Pentium 4, Athlon XP, Athlon 64, and even the Itanium and Itanium 2 have a 64-bit (8-byte) wide data bus. Therefore, they can transfer 64 bits of data at a time to and from the motherboard chipset or system memory.

A good way to understand this flow of information is to consider a highway and the traffic it carries. If a highway has only one lane for each direction of travel, only one car at a time can move in a certain direction. If you want to increase traffic flow, you can add another lane so that twice as many cars pass in a specified time. You can think of an 8-bit chip as being a single-lane highway because 1 byte flows through at a time. (1 byte equals 8 individual bits.) The 16-bit chip, with 2 bytes flowing at a time, resembles a two-lane highway. You might have four lanes in each direction to move a large number of automobiles; this structure corresponds to a 32-bit data bus, which has the capability to move 4 bytes of information at a time. Taking this further, a 64-bit data bus is like having an 8-lane highway moving data in and out of the chip.

Another ramification of the data bus in a chip is that the width of the data bus also defines the size of a bank of memory. So, a processor with a 32-bit data bus reads and writes memory 32 bits at a time, whereas processors with a 64-bit data bus (most current processors) read and write memory 64 bits at a time.

Address Bus

The address bus is the set of wires that carries the addressing information used to describe the memory location to which the data is being sent or from which the data is being retrieved. As with the data bus, each wire in an address bus carries a single bit of information. This single bit is a single digit in the address. The more wires (digits) used in calculating these addresses, the greater the total number of address locations. The size (or width) of the address bus indicates the maximum amount of RAM a chip can address.

The data bus and address bus are independent, and chip designers can use whatever size they want for each. Usually, however, chips with larger data buses have larger address buses. The size of the

data bus is an indication of the chip's information-moving capability, and the size of the address bus tells you how much memory the chip can handle.

Internal Registers (Internal Data Bus)

The size of the internal registers indicates how much information the processor can operate on at one time and how it moves data around internally within the chip. This is sometimes also referred to as the internal data bus. A register is a holding cell within the processor; for example, the processor can add numbers in two different registers, storing the result in a third register. The register size determines the size of data on which the processor can operate. The register size also describes the type of software or commands and instructions a chip can run. That is, processors with 32-bit internal registers can run 32-bit instructions that are processing 32-bit chunks of data, but processors with 16-bit registers can't.

3.1. CPU Removal and Replacement

CPUs are often very easy to remove and replace. Currently used sockets are called zero insertion force (ZIF) sockets. The processor can be gently placed in the socket, and then a lever is lowered and locked to hold the CPU in place. Slot processors should be gently inserted in their slots. Make sure to use proper grounding protection and be careful not to touch the pins.

3.2. Diagnosing CPU Problems

CPUs are generally trouble-free. The most likely problem is heating damage that can happen when the fan wears out. The most common symptom of heat damage is a computer that starts out working normally, but after a while performs erratically. Eventually, the computer will lock up. When hearing a report like this, with the power disconnected, open the case and check to make sure that the fan and heat sink are in place put them in place if they aren't. Attempt to spin the processor fan by hand. The fan should spin freely and continue spinning when you let go, and not make any noise. If you encounter resistance and/or hear noise, replace the fan.

If the fan blades spin freely, the next step is to power on the computer and see if the fan actually spins. Some BIOS setup programs have CPU temperature gauges, and some motherboard manufacturers provide Windows programs to monitor the CPU temperature, among other things. If the fan does spin, you can check these gauges, if present, to determine if the processor's temperature is in the safe zone, which will be indicated on the gauge. Another way to measure the

processor temperature is to use a laser temperature gauge. While the computer is running, aim the laser at the space between the processor and heat. Compare your result to the processor documentation's temperature specifications. If the temperature is too high, the processor isn't being cooled properly.

If there is any problem with the fan spinning, shut down the computer immediately and replace the fan. Then, start up the computer and let it run for at least two hours. Next, open some programs and try to use them. If the computer works normally, you can be reasonably sure that the processor didn't sustain any major damage. If you still have problems, it is possible that the processor has been damaged, or that the computer has other problems.

If the power supply fan spins, but the computer won't otherwise power on, the problem could be a dead or disconnected CPU or motherboard.

Dual-Core Processors

No matter how fast a conventional single-core processor operates or how much RAM is installed in a system, it must ensure that each program and process that is running is properly serviced. As more and more programs are opened, the amount of time the processor can devote to each program is reduced. The result is that system performance declines.

A multithreaded application can run different parts of the program, known as threads, at the same time in the same address space. They can share code and data. Currently, relatively few applications other than video editing programs are multithreaded. Multithreading allows different parts of a program to run concurrently. A multithreaded program runs faster on a dual-core processor than on a single-core processor.

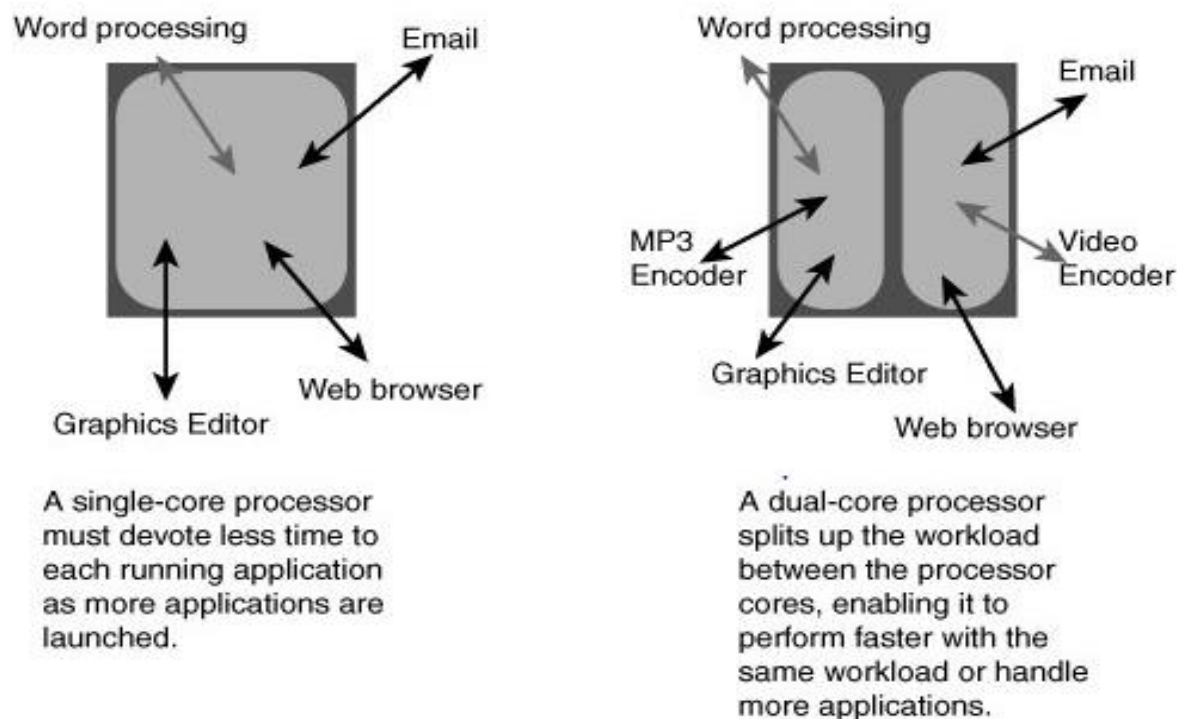
If you use multiple applications at the same time, such as email, web browsers, office suite components such as word processors and spreadsheets, graphics editors, and so forth, you should consider the latest development in processor technology: a dual-core processor. The dual-core processors introduced by Intel and AMD are designed to bring the benefits of multiprocessor operation to desktop systems by placing two processor cores in a single physical processor.

Dual-core processors include two processor cores in the same physical package, providing virtually all the advantages of a multiple-processor computer at a cost lower than that of two matched processors. Dual processor cores provide more time to service each running application or application thread, providing faster performance in a multitasking environment.

Who Needs a Dual-Core Processor?

A dual-core processor is designed for users who frequently multitask (run multiple programs at the same time) or who use multithreaded applications. Figure 1 illustrates how a dual-core processor handles multiple applications for faster performance.

Figure 1 How a single-core processor (left) and a dual-core processor (right) handle multitasking



CPU Fan Replacement Tips

When replacing a fan on a socket processor, make sure you don't slip and damage the motherboard.

Figure 2 shows the steps of installing a processor, fan, and heat sink.

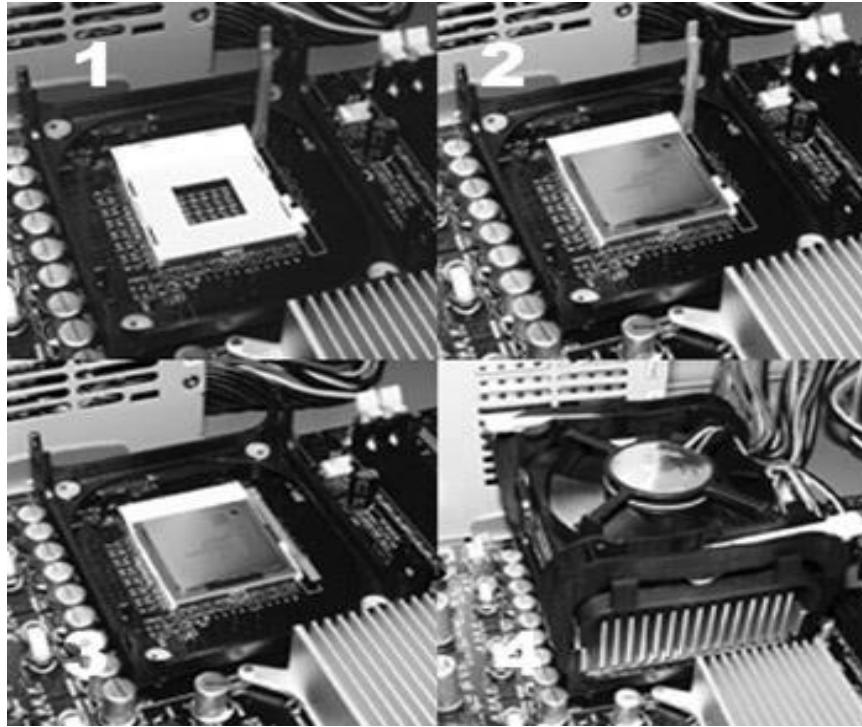


Figure 2: Installing a processor.

Processors are normally very reliable. Most PC problems are with other devices, but if you suspect the processor, there are some steps you can take to troubleshoot it. The easiest thing to do is to replace the microprocessor with a known-good spare. If the problem goes away, the original processor is defective. If the problem persists, the problem is likely elsewhere.

3.3. Types of CPU

CPU or Central Processing Unit is a processor which helps the computer to execute the data and instructions for the user. It processes all the data and instruction is received through the software or hardware which is running on the computer. The CPU contains the circuitry which is needed to process and execute the results. It is known as the CPU (Central Processing Unit) and is also known as the Central Processor which is placed on a motherboard. That's why it is the most important component for a computer and also we know the CPU as the brain of the computer.

Types of CPU (Central Processing Unit)

The Central Processing Unit is categorized by the number of CORES and THREADS. These two things describe how much the CPU is powerful and how much low. The work of the core is to

perform the tasks. These tasks depend on the core (How much core it has) let's take an example CPU A has 1 core and CPU b has 2 cores. So, the CPU A has core 1 it means it perform only one or a minimal task but on the other hand The CPU B has core 2 that mean CPU B perform more tasks than CPU A.

Six different types of CPU are there:

1. Single – Core CPU
2. Dual – Core CPU
3. Quad – Core CPU
4. Hexa – Core CPU
5. Octa – Core CPU
6. Deca – Core CPU

1. Single-Core CPU

The Single Core CPU is that CPU which has core 1 it is known as Single – Core CPU. And its thread is 2. This is the oldest form of CPU which is low power. As I have mentioned that how many cores are there in the CPU, shows their performance (how much they are capable of doing the tasks). In this case, the single-core CPU has only 1 core that means it performs only limited tasks. It's not for multitasking.

But these types of CPU are the very oldest form of CPU Which is introduced in our historical period of time when the CPU was just invented. As such that time this is a very great and powerful CPU But at this time this CPU is not capable to perform our tasks.

The first Single-Core CPU was released by the INTEL Company Model Name = Intel 4004 in 1971.

2. Dual-Core CPU

A Dual-Core CPU is a combination of two Single Core CPUs where it has 2 core and 4 threads. It is the first Multi-Core CPU that is powerful than the single-core CPU. This CPU can easily handle multitasking and make it more comfortable for using and make your experience better than the Single Core.

To make Dual-Core CPU faster in order to run on a fully Optimized Speed. So that it is very important to have a special code in Operating System which is known as SMT (Simultaneous Multi-Thread Technology). It makes the running Program and Applications Smoother and Fast.

The first Dual-Core CPU was also released by INTEL Company Model Name = Intel Pentium D in 2005.

3. Quad-Core CPU

The Quad-Core CPU has Further Improvements than the Dual Core CPU. The word “Quad” means 4, which means it has 4 four core and 8 threads. Which is moreover powerful than the previous core CPU. In Quad Core CPU’s It divides all the work between four CPU Cores. Which makes our experience smoother and performing all the tasks within a second. These types of CPU make it possible to use heavy software like Photo Editing (Photoshop, Illustrator), Video Editing (Premiere Pro), Heavy games Etc.

The first Quad-Core CPU was released by the AMD Company Model Name = Athlon II X4 in 2009.

4. Hexa-Core CPU

The Hexa Core CPU has another level of Performance Which makes our work easier and smoother than the Quad-Core CPU. In this CPU it has 6 core and 12 threads. Complete all the tasks rapidly and also make the faster clock speed, working time, application and software opening time, and performance.

The first Hexa-Core CPU was released by INTEL Company Model Name = Intel core i7-980X in 2010.

5. Octa-Core CPU

The Octa-Core CPU has 8 Core and 16 threads. This means Its performance, application and software workload, clock speed, Rendering Time, etc. and many other things are getting to the next level of performance. Obviously, it is better than the Hexa Core CPU.

The Octa-Core CPU was released by INTEL Company Model Name = Intel Core i7-5960X in 2014

6. Deca-Core CPU

This CPU is the king of performance than the previous Core of CPUs. It has 10 core and 20 Threads. It gives us the killer, super-duper performance. I do not need to tell you about this Deca Core CPU Performance guess yourselves how much this CPU is Powerful.

The Deca- Core was also released by INTEL Company Model Name = Intel Xeon Silver 4114T in 2017.

3.4. CPU Sockets and slots

Slot or socket is the physical connection used to connect a device (CPU) to the system board. The type of slot or socket supplied by the system board for the processor must match that required by the processor.

This is a socket from a Pentium 4 board. These CPUs are fairly easy to install. You basically just drop the CPU into the socket lock it with the locking arm. Then all you have left is to install the cooler and set up the clock speed. This is a slot 1 made for Pentium II, early Pentium III and early Celeron CPUs. It looks a lot like an expansion slot, but is not generally located in the same area on the board. Also once you get the CPU into it with all the cooling it is much bulkier then what you would usually see in any expansion slot.

Following table shows various CPU sockets and slots.

CONNECTOR NAME	USED BY CPU
Socket 4	Classic Pentium 60/66
Socket 5	Classic Pentium 75/90/100/120
Socket 6	Not used
Socket 7	Pentium MMX, Fast Classic Pentium AMD k5, AMD k6, Cyrix M
Socket 8	Pentium Pro
Socket 370 or	Pentium III FC-PGA PGA 370 Socket Celeron PPGA, Cyrix III
Slot 1 or SC242	Pentium II, Pentium III
Slot A	AMD Athalon
Slot 2 or SC330	Pentium II Xeon, Pentium III Xeon

A. ZIF Sockets

Most processors today use special sockets that are designed to let you put the chip in while using "zero insertion force." Instead of having tight-fitting connectors like on a memory chip socket or a slot for a PCI card, these sockets let you practically drop the CPU chip right in. Once it is in place, you rotate a lever that locks the CPU in place. To remove the chip, you just rotate the lever the other way and lift it right out.

B. Ball or Pin

CPU sockets come in two major types -- ball-grid array and pin-grid array. PGA sockets look like a checkerboard with lots of squares. They are designed to hold a CPU chip with an array of pins sticking out of its bottom. BGA and land-grid array sockets, which are usually used in notebook computers and testing applications, are designed to accept CPU chips that do not have pins. BGA sockets frequently require the CPU to be soldered in place.

Pin Numbers and Arrangements

Sockets differ in the number of CPU pins that they can hold. Modern CPU chips transfer 32 or 64 bits of data billions of times per second to the computer's memory, graphics system, storage and other systems, requiring hundreds or thousands of physical connections to support the transfers. As such, if you have an 1155-pin processor, you need an 1155-pin socket. In most cases, you cannot plug a CPU chip with fewer pins into a socket with more pins since the pins that the CPU does have will not line up with either the physical socket or with its internal wiring.

Intel and AMD Sockets

The two major CPU chip makers -- Intel and Advanced Micro Devices -- use different and incompatible sockets. Intel sockets are typically named for the number of pins that they have, so a computer with a Socket 2011 CPU connection holds a 2011-pin CPU. AMD sockets are typically sequentially numbered, with AM- and FM-family sockets both available. Server CPUs and mobile CPUs from both manufacturers also frequently use different sockets from desktop processors.

Chapter Four: Memory

Memory

Memory is the workspace for the computer's processor. It is a temporary storage area where the programs and data being operated on by the processor must reside. Memory storage is considered temporary because the data and programs remain there only as long as the computer has electrical power or is not reset. Before being shut down or reset, any data that has been changed should be saved to a more permanent storage device (usually a hard disk) so it can be reloaded into memory in the future.

When we talk about a computer's memory, we usually mean the RAM or physical memory in the system, which is mainly the memory chips or modules the processor uses to store primary active programs and data. Physically, the main memory in a system is a collection of chips or modules containing chips that are usually plugged into the motherboard. These chips or modules vary in their electrical and physical designs and must be compatible with the system into which they are being installed to function properly.

Three main types of physical memory are used in modern PCs:

-  ROM. Read-only memory
-  DRAM. Dynamic random access memory
-  SRAM. Static RAM

ROM

Read-only memory, or ROM, is a type of memory that can permanently or semi permanently store data. It is called read-only because it is either impossible or difficult to write to. ROM also is often referred to as nonvolatile memory because any data stored in ROM remains there, even if the power is turned off. As such, ROM is an ideal place to put the PC's startup instructions that is, the software that boots the system.

DRAM

Dynamic RAM (DRAM) is the type of memory chip used for most of the main memory in a modern PC. The main advantages of DRAM are that it is very dense, meaning you can pack a lot of bits into a very small chip, and it is inexpensive, which makes purchasing large amounts of memory affordable.

The memory cells in a DRAM chip are tiny capacitors that retain a charge to indicate a bit. The problem with DRAM is that it is dynamic. Also, because of the design, it must be constantly refreshed; otherwise, the electrical charges in the individual memory capacitors will drain and the data will be lost. Refresh occurs when the system memory controller takes a tiny break and accesses all the rows of data in the memory chips. Most systems have a memory controller, which is set for an industry-standard refresh time of 15ms (milliseconds). This means that every 15ms, all the rows in the memory are automatically read to refresh the data.

Refreshing the memory unfortunately takes processor time away from other tasks because each refresh cycle takes several CPU cycles to complete. In older systems, the refresh cycling could take up to 10% or more of the total CPU time, but with modern systems running in the multi-gigahertz range, refresh overhead is now on the order of a fraction of a percent or less of the total CPU time. Some systems allow you to alter the refresh timing parameters via the CMOS Setup. The time between refresh cycles is known as tREF and is expressed not in milliseconds, but in clock cycles. Figure 1 shows that the refresh period dialog box and other advanced memory timings can be adjusted manually through the system CMOS setup program.

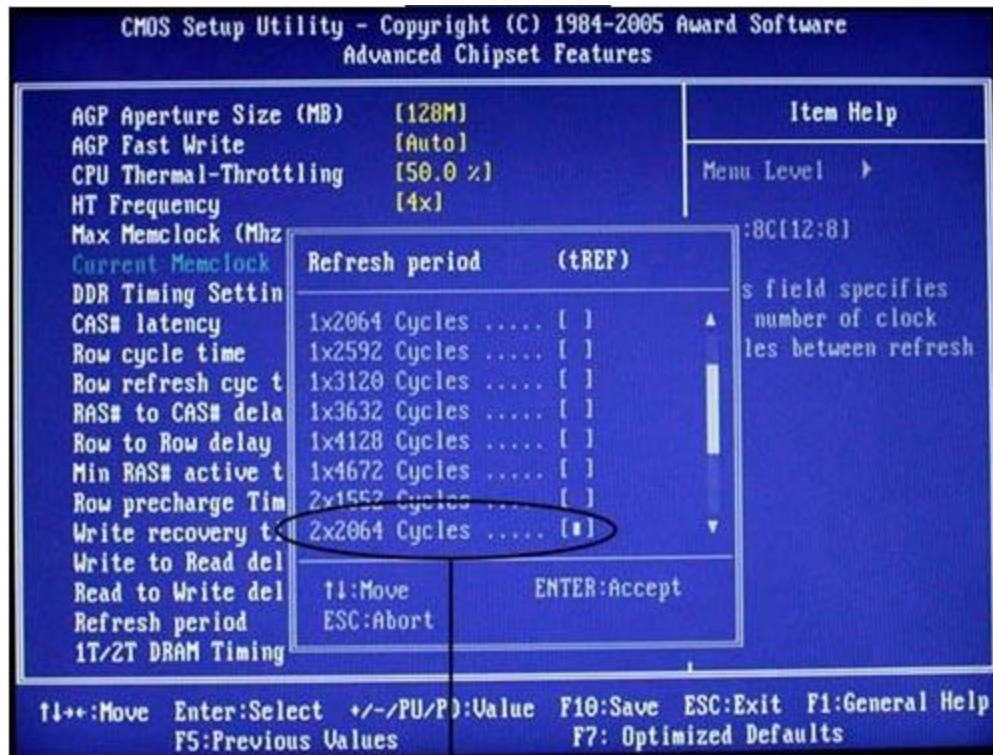


Figure 1: Current tREF (refresh period) for the motherboard

It's important to be aware that increasing the time between refresh cycles (tREF) to speed up your system can allow some of the memory cells to begin draining prematurely, which can cause random soft memory errors to appear.

A soft error is a data error that is not caused by a defective chip. To avoid soft errors, it is usually safer to stick with the recommended or default refresh timing. Because refresh consumes less than 1% of modern system overall bandwidth, altering the refresh rate has little effect on performance. It is almost always best to use default or automatic settings for any memory timings in the BIOS Setup. Many modern systems don't allow changes to memory timings and are permanently set to automatic settings.

The transistor for each DRAM bit cell reads the charge state of the adjacent capacitor. If the capacitor is charged, the cell is read to contain a 1; no charge indicates a 0. The charge in the tiny capacitors is constantly draining, which is why the memory must be refreshed constantly. Even a momentary power interruption, or anything that interferes with the refresh cycles, can cause a DRAM memory cell to lose the charge and therefore the data. If this happens in a running system,

it can lead to blue screens, global protection faults, corrupted files, and any number of system crashes.

DRAM is used in PC systems because it is inexpensive and the chips can be densely packed, so a lot of memory capacity can fit in a small space. Unfortunately, DRAM is also slow, typically much slower than the processor. For this reason, many types of DRAM architectures have been developed to improve performance.

Cache Memory: SRAM

Another distinctly different type of memory exists that is significantly faster than most types of DRAM. SRAM stands for static RAM, which is so named because it does not need the periodic refresh rates like DRAM. Because of how SRAMs are designed, not only are refresh rates unnecessary, but SRAM is much faster than DRAM and much more capable of keeping pace with modern processors.

Compared to DRAM, SRAM is much faster but also much lower in density and much more expensive (see [Table 1](#)). The lower density means that SRAM chips are physically larger and store fewer bits overall. The high number of transistors and the clustered design mean that SRAM chips are both physically larger and much more expensive to produce than DRAM chips. For example, a DRAM module might contain 64MB of RAM or more, whereas SRAM modules of the same approximate physical size would have room for only 2MB or so of data and would cost the same as the 64MB DRAM module. Basically, SRAM is up to 30 times larger physically and up to 30 times more expensive than DRAM. The high cost and physical constraints have prevented SRAM from being used as the main memory for PC systems.

Table 1: Comparing DRAM and SRAM

Type	Speed	Density	Cost
DRAM	Slow	High	Low
SRAM	Fast	Low	High

Types of Memory Chips

Not only do you have to specify what type of memory module your system uses, you also have to specify what type of RAM chip is installed on the module. There are different primary types of memory chips in use today: SDRAM and DDR SDRAM are types of memory chips.

SDRAM

SDRAM is short for synchronous DRAM, a type of DRAM that runs in synchronization with the memory bus. SDRAM delivers information in very high-speed bursts using a high-speed, clocked interface. SDRAM removes most of the latency (the time it takes for a specific block of data to move) involved in asynchronous DRAM because the signals are already in synchronization with the motherboard clock.

DDR SDRAM

Double data rate (DDR) SDRAM memory is an evolutionary upgrade of standard SDRAM in which data is transferred twice as quickly. Instead of doubling the actual clock rate, DDR memory achieves the doubling in performance by transferring twice per transfer cycle: This effectively doubles the transfer rate, even though the same overall clock and timing signals are used.

Types of RAM Modules

All memory today comes on modules that contain multiple memory chips; the capacity of each chip adds up to the total capacity of the memory module. The memory modules plug into memory sockets located on your PC's motherboard; installation is actually quite easy.

Your computer could be using one of three different types of memory modules—SIMMs, DIMMs, or RIMMs. You'll need to consult your PC's instruction manual to determine the type of module your machine uses. See Figure 2 for a comparison of the various types of memory modules.

Figure 2: Most systems in production today use DIMMs, though you will find some that use RIMMs; SIMMs are found only in older computers.

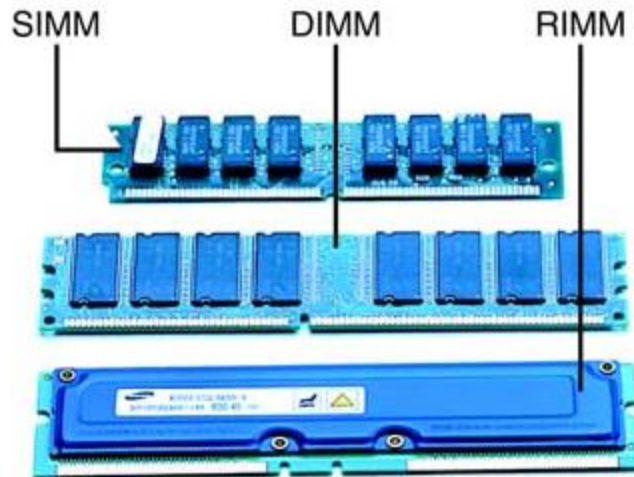


Figure 2: Types of RAM Modules

SIMM

SIMM stands for Single Inline Memory Module. Chips were originally mounted on only one side of a SIMM (hence the "single" in the name), although newer modules have chips on both sides. SIMMs use either 30-pin or 72-pin connectors.

DIMM

DIMM stands for Dual Inline Memory Module. Unlike single-sided SIMMs, DIMMs have memory chips mounted on both sides of the module. They come in both 168-pin and 184-pin versions. (DIMMs are the most popular memory modules in use today.)

Note: Although it's important to get the right type of module for your specific PC. It's physically impossible to insert a module into a slot that doesn't match it, pin-wise—so you can't insert a 184-pin RIMM into a 232-pin slot

RIMM

RIMM stands for Rambus Inline Memory Module. RIMMs use 184-pin or 232-pin connectors and are slightly faster than comparable DIMMs. RIMMs also cost at least twice as much DIMMs; for that reason, they're not widely used in most consumer-oriented PCs.

4.1. Memory Installation

Memory isn't hard to install. First, make sure to take the usual static-avoidance measures, including wearing an anti-static wrist strap. Touch a piece of bare metal on the case before proceeding. With DIMMs, insert the module straight in the slot, taking care to match up the indentation in the module with the notch in the slot (see Figure 3). Then, make sure the plastic clips are closed as shown in the Figure below.

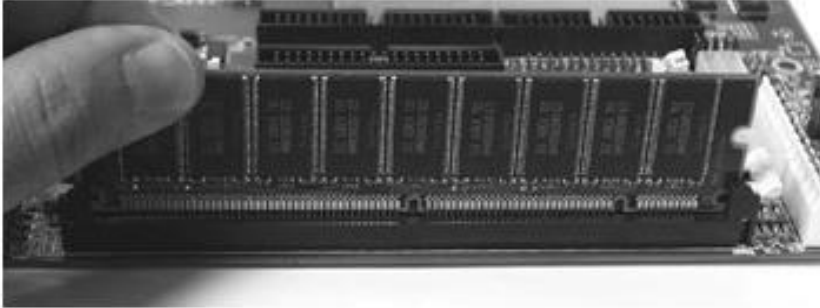


Figure 3: Match the indentation with the notch.

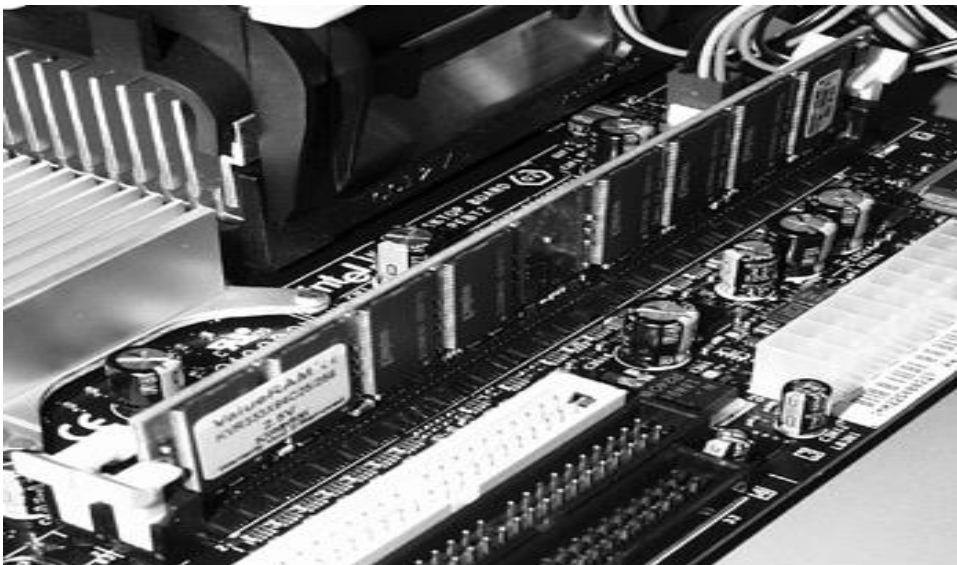


Figure 4: An installed module.

Upgrade: Adding RAM to a Desktop PC

Just follow these steps:

1. Close Windows and power off your PC.
2. Disconnect your PC from its power source.
3. Open the system unit case.

4. Locate your system's empty memory slots.
5. Plug the new memory module into an open memory slot.
6. Reinstall the system cover, reconnect your PC to an AC power supply, and power it

Caution: Like any other types of devices, it's important to protect against electrostatic discharge while you're installing memory. Use an antistatic wrist band or take other precautions to make sure you don't fry your new RAM before you get to use it.

Most newer PCs will automatically recognize the new additional memory when you reboot your PC. If your system displays any sort of error message when you start it up, you'll need to enter the BIOS setup program and reconfigure your system settings manually for the proper amount of memory now installed.

Note that each type of module inserts into its slot a little differently, as you can see in Figure 5. You release the clips at either end of the slot, line up the module, press it firmly down into the slot, and then flip the end clips back into place.

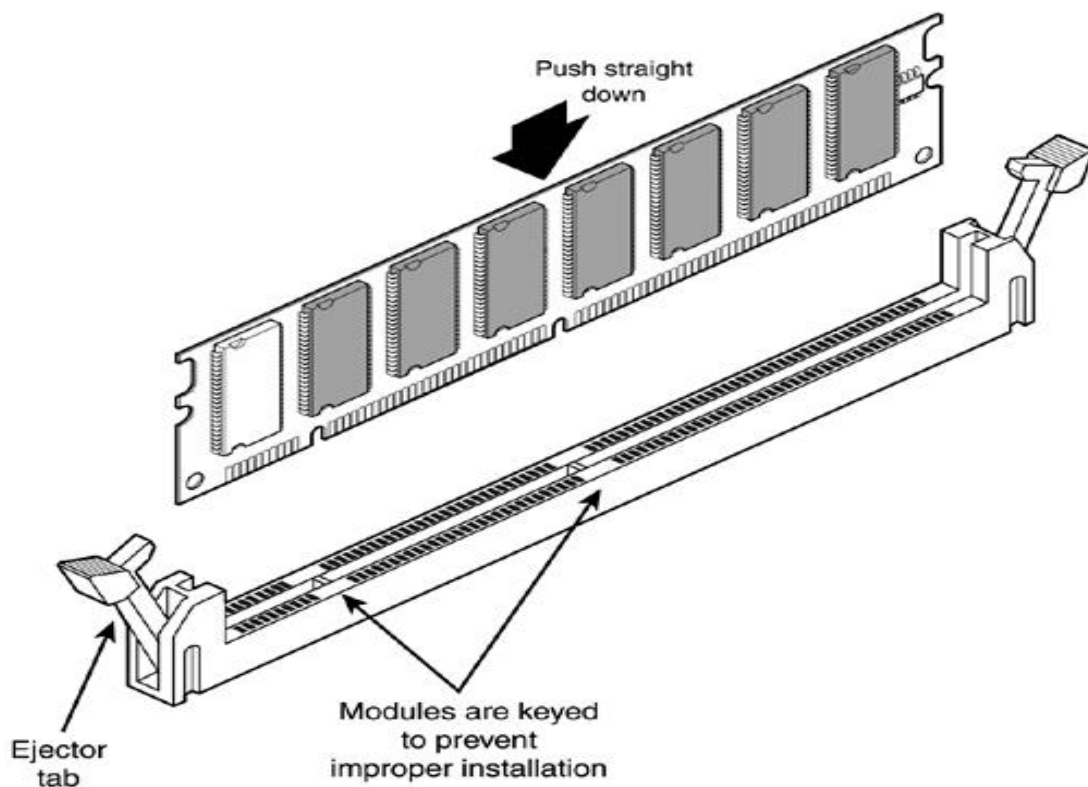


Figure 5: Inserting a DIMM module.

4.2. Types of memory

Memory is the electronic holding place for the instructions and data a computer needs to reach quickly. It's where information is stored for immediate use. Memory is one of the basic functions of a computer, because without it, a computer would not be able to function properly. Memory is also used by a computer's operating system, hardware and software.

There are technically two types of computer memory: primary and secondary. The term *memory* is used as a synonym for primary memory or as an abbreviation for a specific type of primary memory called random access memory (RAM). This type of memory is located on microchips that are physically close to a computer's microprocessor.

If a computer's central processor (CPU) had to only use a secondary storage device, computers would become much slower. In general, the more memory (primary memory) a computing device has, the less frequently the computer must access instructions and data from slower (secondary) forms of storage.

Chapter 5: Power Supplies

5.1. Power supplies

A power supply is a hardware component that supplies power to an electrical device. It receives power from an electrical outlet and converts the current from AC (alternating current) to DC (direct current), which is what the computer requires. It also regulates the voltage to an adequate amount, which allows the computer to run smoothly without overheating. The power supply is an integral part of any computer and must function correctly for the rest of the components to work.

Power Supply

The basic function of the power supply is to convert the type of electrical power available at the wall socket to the type the computer circuitry can use. The power supply in a conventional desktop system is designed to convert either 120-volt AC (alternating current) or 240V AC power into +3.3V, +5V, and +12V DC (direct current) power.

DC Power Output Connectors

The DC power connectors vary somewhat between AT, ATX, and proprietary form factors. While the disk drive connectors are all the same, the motherboard connectors are different and incompatible between major form factors. Figure 10 shows different ATX DC power connectors

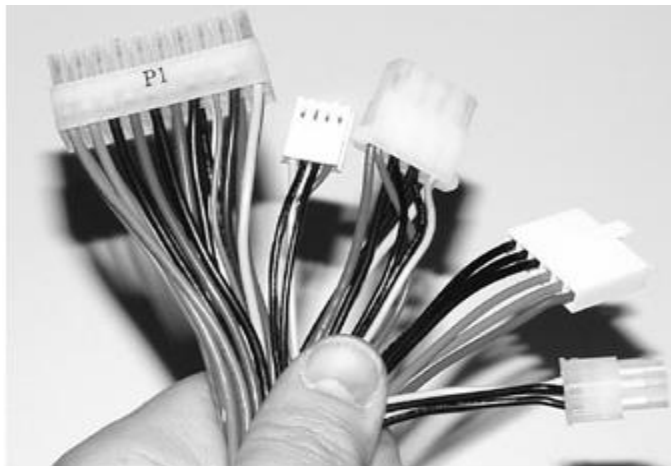


Figure 10: ATX DC power connectors

Diagnosing Power Supply Problems

The most obvious symptoms of a failed power supply occur when you turn your computer on and hear a loud noise, smell something burning, and/or see smoke, and the computer won't power on.

Shut off the power immediately; you likely have a blown power supply. If there is smoke, you'll want to ventilate the room—it is toxic. Usually, however, the symptoms of a blown power supply aren't nearly so dramatic; you turn on the computer and nothing happens.

ATX Main Power Connector

The industry-standard ATX power supply-to-motherboard main connector is the Molex. This is more commonly known as the 20-pin ATX-style connector see Figure 11.

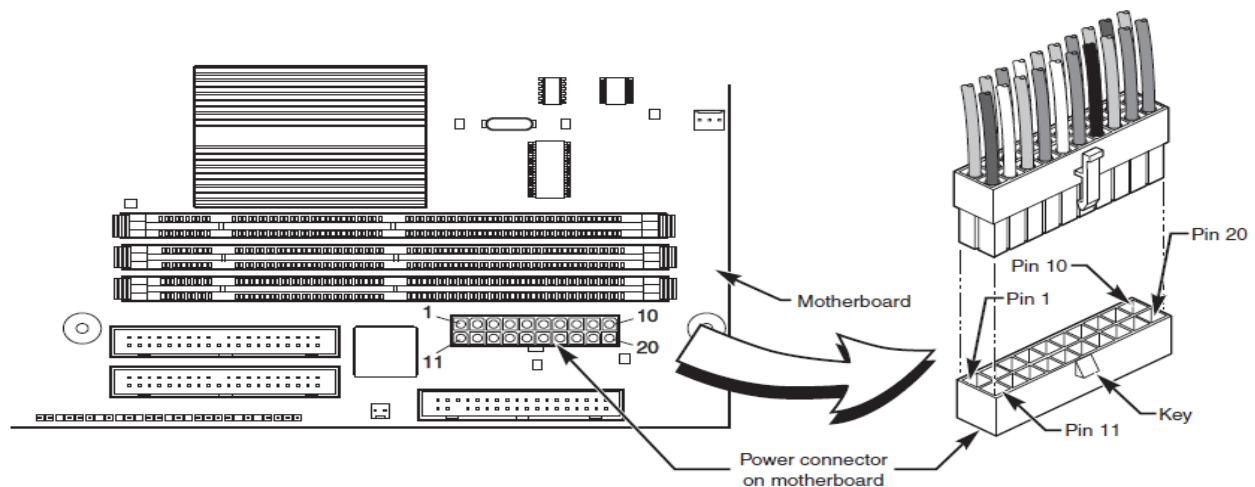


Figure 11: ATX-style 20-pin motherboard main power connector

Installing a Power Supply

Standard power supplies are secured with four screws. To remove a power supply, disconnect the AC supply cord, the motherboard power cable(s), and all device power cables. Use one hand to hold the power supply in place while removing the four screws that secure it, and then lift it straight out. Some power supplies use a locking tab and slot arrangement, so you may have to slide the power supply a short distance to clear the tab before lifting it out. To install a power supply, reverse that process. Slide the power supply into place, as shown in Figure 12, making sure that the locking tab, if present, mates with the slot.



Figure 12: Slide the power supply into position

Once the power supply is in place, align the screw holes and insert the screws, as shown in Figure 13. If necessary, support the power supply with one hand while you insert screws with the other. Most good cases have a tray that supports the power supply, while other cases simply leave the power supply hanging in mid-air, secured only by the screws. In the latter situation, you may want to get someone to volunteer a second pair of hands to hold the power supply while you insert the screws, particularly if you're working in an awkward position. A motherboard can be damaged if the power supply drops on it.



Figure 13: After aligning the power supply within the chassis, secure it by inserting the four screws

Once the power supply is in place and secured, check the position of the voltage selection switch, shown in Figure 14, and reset it if necessary. Do not neglect this step. If the power supply voltage switch is set to 220/240V and you connect the system to 110/120V mains, nothing bad happens, though the system doesn't run. But if your mains voltage is 220/240V and you leave the power supply voltage switch set to 110/120V, you can expect your motherboard and connected components to literally burn to a crisp within a second of applying power to it. Some power supplies are autosensing and so do not have a voltage selection switch. But if yours does have such a switch, make sure to set it properly before you connect any cables.

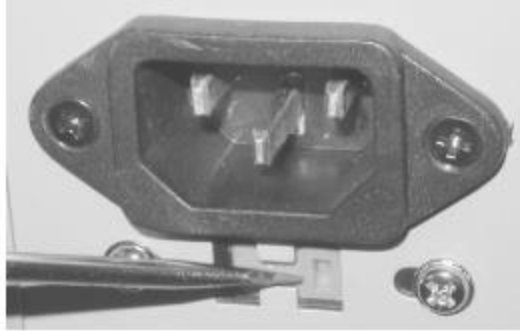


Figure 14: Set the voltage selection switch for the correct local voltage

Once you have installed and secured the power supply and set the voltage selection switch, reconnect the motherboard power cables and all drive power cables. If your system has auxiliary fans that are powered from power supply connectors, reconnect those too. Once you've verified that everything is installed and connected correctly, reconnect the main power cable and apply power to the system.

5.2. Troubleshooting the Power Supply

Suspect a power supply problem if you experience any of the following symptoms, particularly in combination:

- ✚ Parity check errors. Such errors may be caused by defective or poorly seated memory or by overheating, but insufficient or poorly regulated +3.3VDC or +5VDC from a failing or inadequate power supply is a likely cause.
- ✚ Sporadic or regular boot failures. Obviously, such errors may instead be caused by hard disk, cable, or disk controller problems, but inadequate or poorly regulated +12VDC (less commonly, +5VDC) is also a common cause of this problem.
- ✚ Spontaneous reboots or system lockups during routine operations, not attributable to running a particular program. Numerous other factors can cause this problem, but one common cause is insufficient or poorly regulated +3.3VDC and/or +5VDC being provided to the memory and/or processor.
- ✚ Slow disk performance. Although this may seem an odd symptom to be related to power supply problems, inadequate voltage and current can cause disk retries on both reads and writes. The error correction circuitry built into hard disks and controllers means that this problem often (usually) goes undiagnosed. People often say to us something like, "I

replaced the power supply as you suggested, and now my hard disk seems a lot faster. Is that possible?" Yes, it is.

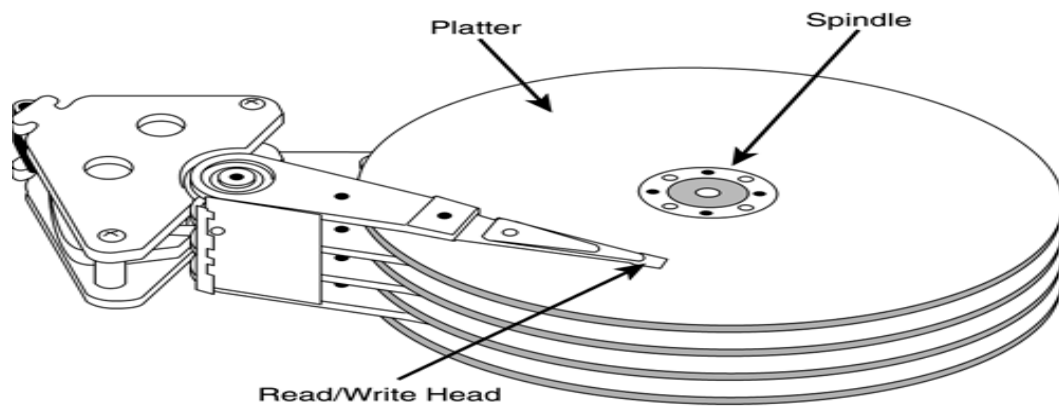
Chapter 6: Storage Devices

6.1. Hard Disk

A hard disk drive is a sealed unit that a PC uses for nonvolatile data storage. Nonvolatile, or semi-permanent, storage means that the storage device retains the data even when no power is supplied to the computer. Because the hard disk drive is expected to retain data until deliberately erased or overwritten, the hard drive is used to store crucial programming and data. As a result, when the hard disk fails, the consequences are usually very serious. To maintain, service, and upgrade a PC system properly, you must understand how the hard disk functions. A hard drive consists of

multiple rigid-metal platters, stacked on a spindle inside a sealed enclosure. The spindle is attached to a motor that spins the platters at very high speeds.

Figure 1: The inside of a hard drive looks like this



Hard disk platters are coated with a special material that enables the storage of information in magnetic patterns. Data is written to and read from the platter by means of an electromagnetic read/write head. Each platter is divided into sectors; each sector contains 512 bytes of data.

Access Time

Access time is the amount of time it takes for the heads to locate a specific piece of data on the hard drive. Manufacturers typically specify the "average access time," because the actual seek time varies depending on the location of the heads and where the next bit of data is stored.

Spin Rate

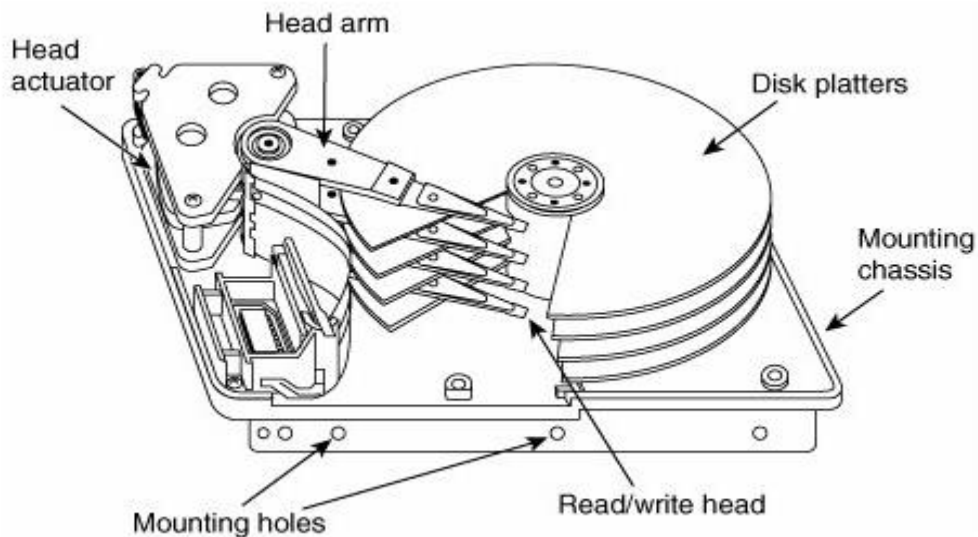
This spec measures the speed at which the platters spin, in revolutions per minute (RPM). Faster-spinning drives result in faster data transfer rates; most drives today spin at either 5,400 RPM or 7,200 RPM. If you're doing a lot of audio or video-related tasks, you'll want a faster drive.

Data Transfer Rate

The data transfer rate is the speed at which the system copies data from the hard drive to your computer (and vice versa).

A hard disk drive contains rigid, disk-shaped platters. Unlike floppy disks, the platters can't bend or flex hence the term hard disk. In most hard disk drives, you can't remove the platters, which is why they are sometimes called fixed disk drives.

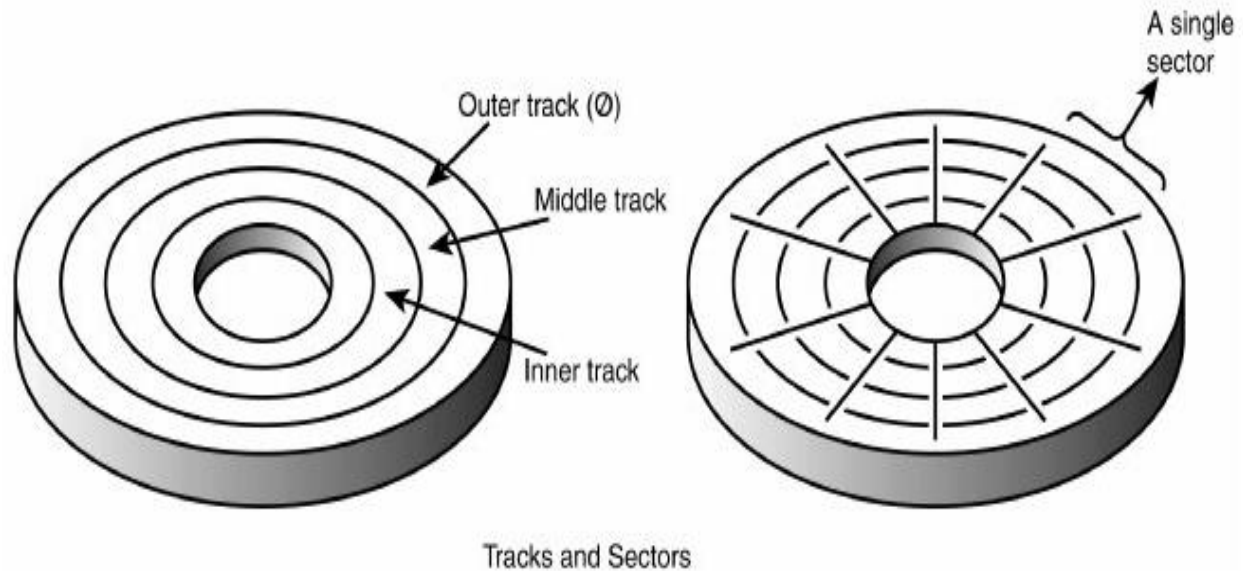
Figure 2: Hard disk heads and platters



Hard Disk Drive Operation

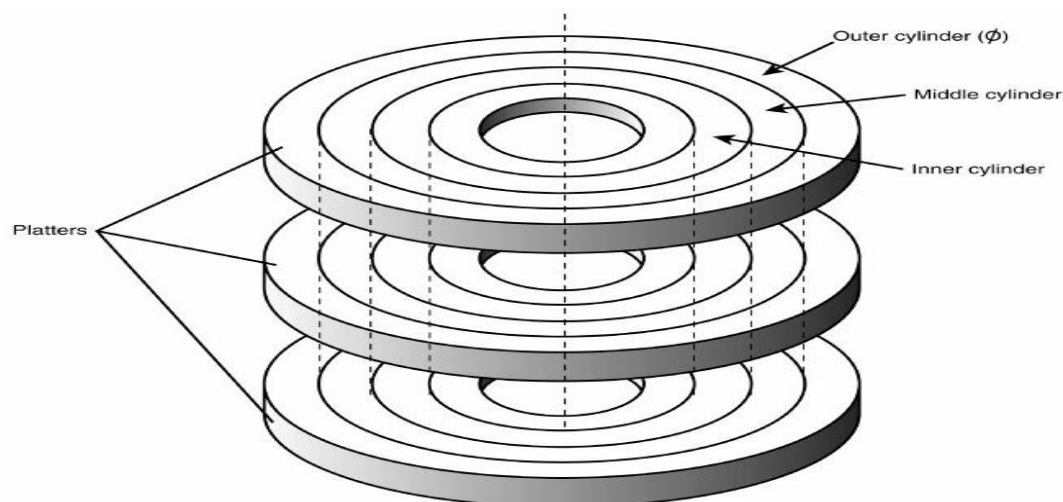
The basic physical construction of a hard disk drive consists of spinning disks with heads that move over the disks and store data in tracks and sectors. The heads read and write data in concentric rings called tracks, which are divided into segments called sectors, which typically store 512 bytes each (see Figure 3).

Figure 3: The tracks and sectors on a disk



Hard disk drives usually have multiple disks, called platters, that are stacked on top of each other and spin in unison (at the same time), each with two sides on which the drive stores data. Most drives have two or three platters, resulting in four or six sides, but some PC hard disks have up to 12 platters and 24 sides with 24 heads to read them. The identically aligned tracks on each side of every platter together make up a cylinder (see Figure 4). A hard disk drive usually has one head per platter side, with all the heads mounted on a common carrier device or rack. The heads move radially across the disk in unison; they can't move independently because they are mounted on the same carrier or rack, called an actuator.

Figure 4: Hard disk cylinders



Originally, most hard disks spun at 3,600rpm approximately 10 times faster than a floppy disk drive. For many years, 3,600rpm was pretty much a constant among hard drives. Now, however, most drives spin even faster. Although speeds can vary, modern drives typically spin the platters at either 4,200rpm; 5,400rpm; 7,200rpm; 10,000rpm; or 15,000rpm. Most standard-issue drives found in PCs today spin at 7,200rpm, with high-performance models spinning at 10,000rpm, although many less-expensive drives still spin at 5,400rpm. 15,000rpm drives are usually found only in very high-performance workstations or servers, where their higher prices. High rotational speeds combined with a fast head-positioning mechanism and more sectors per track are what make one hard disk faster overall than another.

Tracks and Sectors

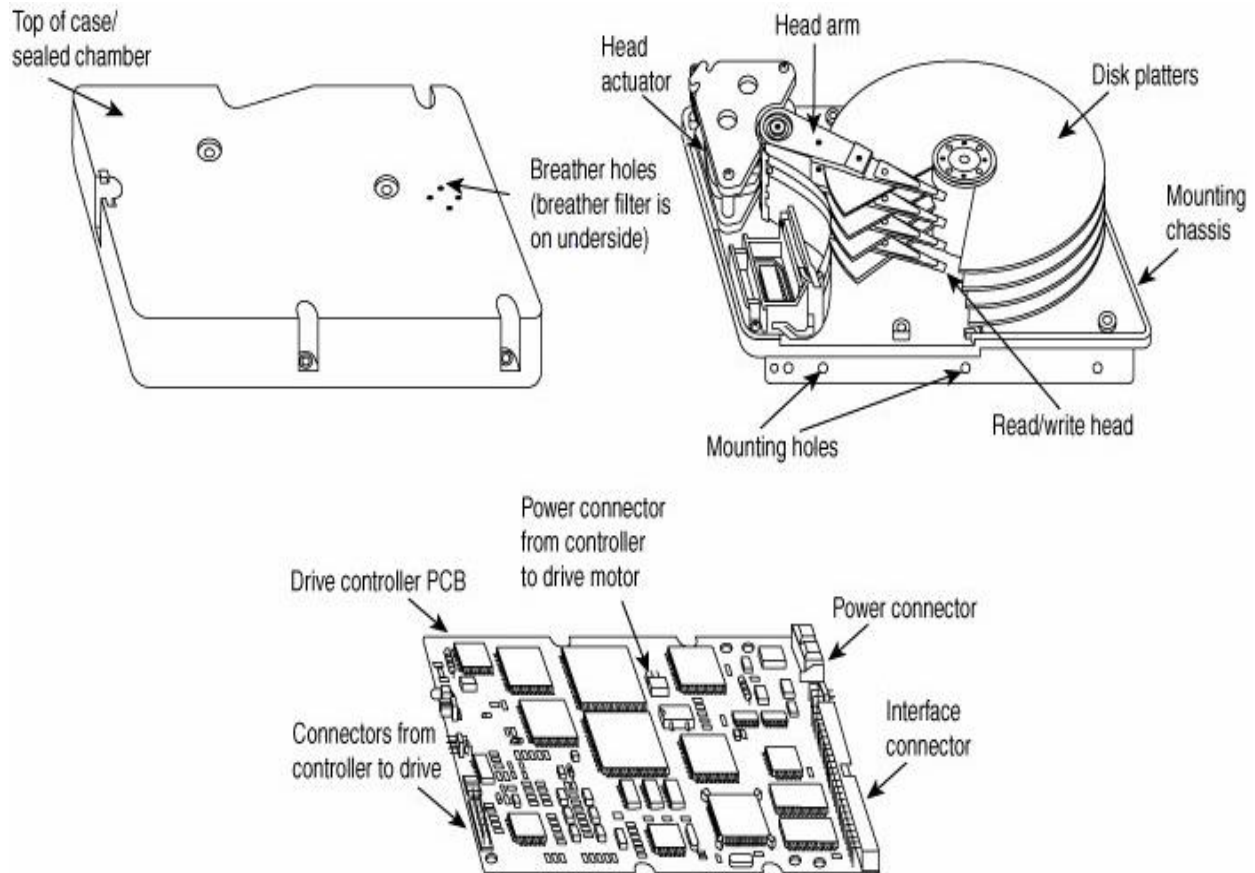
A track is a single ring of data on one side of a disk. A disk track is too large to manage data effectively as a single storage unit. Many disk tracks can store 100,000 or more bytes of data, which would be very inefficient for storing small files. For that reason, tracks are divided into several numbered divisions known as sectors. These sectors represent arc-shaped pieces of the track. The sectors created by the standard formatting procedure on a PC system have a capacity of 512 bytes, which has been one constant throughout the history of the PC.

Basic Hard Disk Drive Components

Many types of hard disk drives are on the market, but nearly all share the same basic physical components. Some differences might exist in the implementation of these components (and in the quality of the materials used to make them), but the operational characteristics of most drives are similar. The basic components of a typical hard disk drive are as follows (see Figure 5):

-  Disk platters
-  Read/write heads
-  Head actuator mechanism
-  Spindle motor (inside platter hub)
-  Logic board (controller or Printed Circuit Board)
-  Cables and connectors
-  Configuration items (such as jumpers or switches)

Figure 5: Typical hard disk drive components



6.2.Disk Formatting

Two formatting procedures are required before you can write user data to a disk:

- ✚ **Physical, or low-level formatting**

- ✚ **Logical, or high-level formatting**

A hard disk, however, requires two separate formatting operations. Moreover, a hard disk requires a third step, between the two formatting procedures, to write the partitioning information to the disk. Partitioning is required because a hard disk is designed to be used with more than one operating system. Using multiple operating systems on one hard drive is possible by separating the physical formatting in a procedure that is always the same, regardless of the operating system used and the high-level format (which is different for each operating system). Partitioning enables a single hard disk drive to run more than one type of operating system, or it can enable a single operating system to use the disk as several volumes or logical drives. A volume or logical drive is any section of the disk to which the operating system assigns a drive letter or name.

Consequently, preparing a hard disk drive for data storage involves three steps:

1. **Low-level formatting (LLF)**
2. **Partitioning**
3. **High-level formatting (HLF)**

Low-Level Formatting

During a low-level format, the formatting program divides the disk's tracks into a specific number of sectors, creating the inter sector and inter track gaps and recording the sector header and trailer information. The program also fills each sector's data area with a dummy (having the appearance of being real but lacking capacity to function) byte value or a pattern of test values. For hard disks, the number of sectors per track depends on the drive and the controller interface.

Partitioning

Creating a partition on a hard disk drive enables it to support separate file systems, each in its own partition.

Each file system can then use its own method to allocate file space in logical units called clusters or allocation units. Every hard disk drive must have at least one partition on it and if it has more than one partition, each of which can support the same or different type file systems. Three common file systems are used by PC operating systems today:

- ✚ FAT (file allocation table). The standard file system supported by DOS and Windows 9x/Me. FAT partitions support filenames of 11 characters maximum (8 characters + a 3-character extension) under DOS, and 255 characters under Windows 9x (or later). The standard FAT file system uses 12- or 16-bit numbers to identify clusters, resulting in a maximum volume size of 2GB.

- ✚ FAT32 (file allocation table, 32-bit). An optional file system supported by Windows 95, Windows 98, Windows Me, and Windows 2000/XP.

FAT32 uses 32-bit numbers to identify clusters, resulting in a maximum single volume size of 2TB or 2,048GB.

- ✚ NTFS (Windows NT File System). The native file system for Windows NT/2000/XP that supports filenames up to 256 characters long and partitions up to (a theoretical) 16 exabytes

(A unit of information equal to 1024 pebibytes or 2^{60} bytes). NTFS also provides extended attributes and file system security features that do not exist in the FAT file system.

Up until the release of XP, FAT32 was by far the most popular file system. Because NTFS is native to XP, NTFS is now more popular in newer systems. FAT32 and NTFS provide additional features but are not universally accessible by other operating systems.

High-Level Formatting

During the high-level format, the operating system writes the structures necessary for managing files and data on the disk. For example, FAT partitions have a Volume Boot Sector (VBS), two copies of a file allocation table (FAT), and a root directory on each formatted logical drive. These data structures enable the operating system to manage the space on the disk, keep track of files, and even manage defective areas so they do not cause problems.

High-level formatting is not really a physical formatting of the drive, but rather the creation of a table of contents for the disk. In low-level formatting, which is the real physical formatting process, tracks and sectors are written on the disk.

6.3. Floppy Drive

A floppy disk drive (FDD), or floppy drive, is a hardware device that reads data storage information. It was invented in 1967 by a team at IBM and was one of the first types of hardware storage that could read/write a portable device. FDDs are used for reading and writing on removable floppy discs. Floppy disks are now outdated, and have been replaced by other storage devices such as USB and network file transfer.

A floppy disk commonly came in three sizes, 8 inches, 5.5 inches and 3.5 inches, becoming smaller as the technology advanced. The newer, 3.5-inch version used more cutting-edge technology and held more data than previous models, while the original 8-inch floppy drive was developed to load hardware-level instructions and/or data structures called microcode into the IBM System/370 mainframe. The 8-inch flexible diskette was read-only, held 80 kilobytes of memory and was referred to as a memory disk. Eight-inch floppy drives did not connect to the motherboard, but rotated on a turntable that was run by an idler wheel.

As the floppy disk advanced to a smaller 5.5- and 3.5-inch designs, the FDD changed as well. To accommodate a smaller floppy disk, an FDD had to make aggressive changes by matching the size of the floppy disk drive opening to the size of the floppy disk for compatibility. For many years, the majority of PCs and notebooks had a floppy drive. Using a floppy disk to exchange data between PCs was a standard method for many computer technicians. The floppy disk was one of the most common ways to store adequate amounts of data outside of a computer's hard drive for personal use because they were inexpensive and easy to carry.

As technology advanced, floppy disks were finally able to read and write. By this point, FDDs had four basic components:

1. Magnetic read/write heads (one or two)
2. A spindle clamping device that held the disk in place as it was spinning 300 to 360 rotations per minute
3. A frame with levers that opened and closed the device
4. A circuit board that contained all of the electronics.

The read/write heads could read both sides of a disk, and the same head was used for reading and writing. A separate, wider head was used for erasing data to ensure that all data was erased without having to interfere with the data already on the adjacent track.

A floppy drive cable could house two drives. In a computer system, the drive at the end of the cable was drive A. When another drive was added, it was connected to the middle of the cable and was called drive B.

Chapter 7: Bus and Cards

7.1. Bus

A bus in computers is a digital pathway that connects components of the computer. Early in computer development, components were connected by cables and bundles, which make up a bus in electrical terms.

Since the discrete components must communicate, the bus is the conduit on which that communication travels. In modern computers, it is a high-speed internal communication connection. The system bus can be viewed as the internal communication transport system for the computer.

System Bus

The system bus is the connection between the main groups of components. The main hardware components of a computer system are the central processing unit (CPU), the memory, and the input/output (I/O) devices. The system bus is also referred to as the CPU bus and as the front side bus. The back side bus connects to the CPU memory cache and is generally internal to the CPU chip in modern computers. The purpose of the system bus is data transfer between the CPU, Memory and I/O Devices.

The CPU does all the mathematical computations and processing of data. The memory is where the data is stored to be used by the CPU and where the results of the processing is stored. The I/O devices have the function of interacting with other components, systems and eventually, with the user.

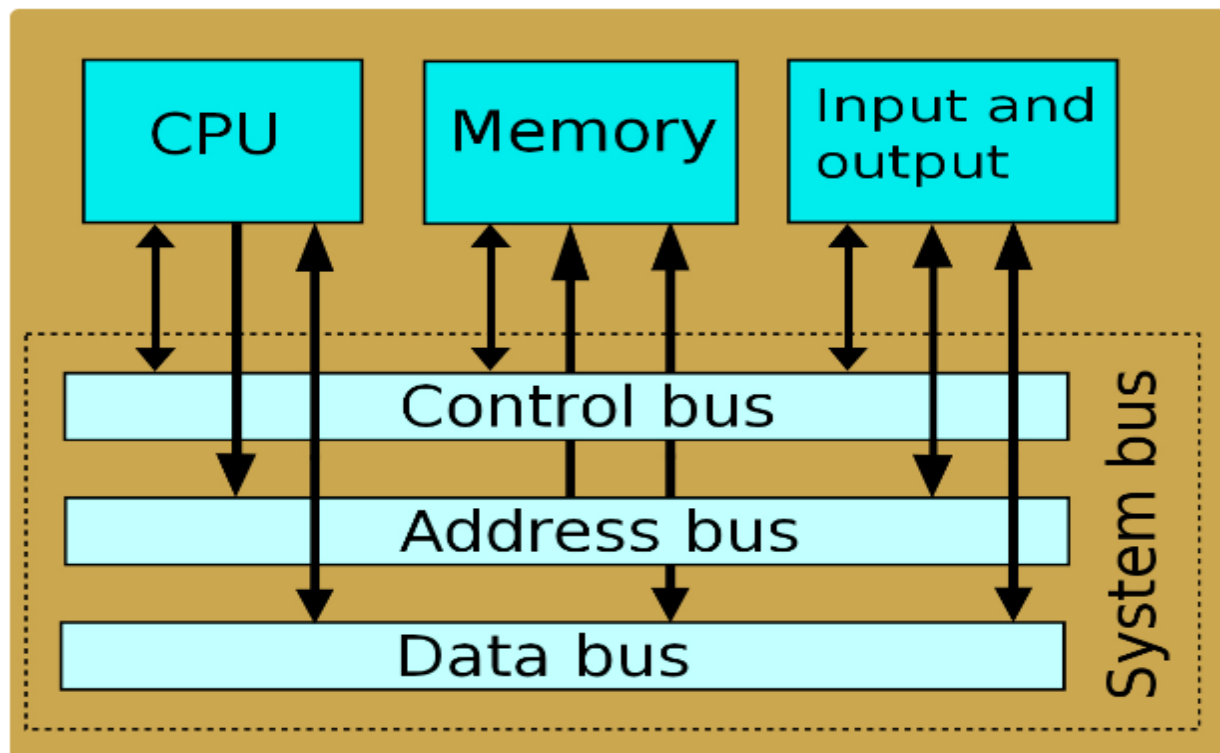
The CPU places a data request on the bus using an address or location in the memory or I/O device. The next component reads the request and then places the data on the bus to return to the CPU, which then reads the data request for processing. Each of these actions are done on a specific control cycle. How fast the cycle occurs is the system bus speed. All the components on the cpu bus operate at the same speed. A higher system bus speed will allow the computer to process data quicker. The bus speed is measured as a frequency in the number of cycles per second using the unit of hertz.

How much data can be simultaneously transferred with each cycle is the system bus width. A wider bus width allows more data to be transferred per cycle, resulting in the computer being able to

process data faster. The bus width is measured in the number of bits. A bit is a discrete one or zero state.

The system bus connects the CPU with the other components, like memory storage and I/O devices, to facilitate communications between those components. This is all done through the physical wiring etched on the motherboard. The system bus is also called the front side bus. There are other types of buses in the computer, including the back-side bus, the peripheral bus and the Universal Serial Bus (USB).

Peripherals are not connected directly to the system bus. Instead, there is a peripheral bus for these devices to directly connect. The peripheral bus is then connected to the system bus. Peripherals can have different data transfer speeds and formats, as well as varying types of control logic, so a peripheral bus is used to bridge to the system bus. Higher speed devices and lower speed devices are commonly bridged differently to maximize operational efficiency.



7.2. Types of Bus

There are three types of buses, each operating separately with its own purpose and function, that make up a system bus. Each type of bus follows a unique set of instructions that contribute to the accurate transfer of data. These three types include:

Data bus: The data bus carries the actual data between the computer's main processor, the memory and all other internal hardware components. It operates between computer parts or devices to increase the flow of data by transferring data in or out of the CPU or from one device to another.

Address bus: The address bus tells the system where data comes from or goes. Typically, this bus takes data from the central processor and stores it in a specific location in the computer's memory, or vice versa.

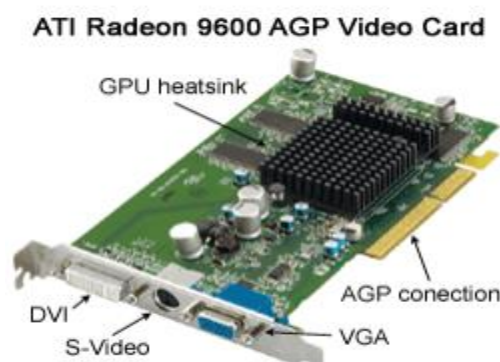
Control bus: The control bus determines how the system bus operates by managing the control, timing and coordination of the busses to ensure the data transfers without corruption. This bus is important for accuracy and security, as it safeguards data during the transfer.

7.3. Cards

Different types of cards used in a computer. In a computer many different cards are used. You may have heard of their names quite often. Video card, Sound card, network interface card, Ethernet card, Accelerator card, Video capture card, TV Tuner card are some of the cards used in a computer. Some of them are essential cards while some others are inserted if needed. Generally, any computer will have a sound card and a video card. In recent computers the network card also is very common. Here let us define the use of these cards:

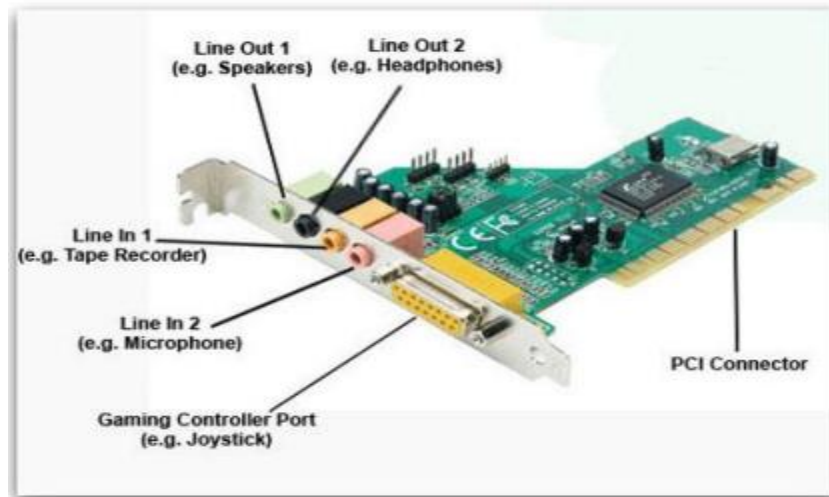
7.4. Video Card

This is also referred to as the graphics adapter, display adapter or video adapter. This is a circuit that is used to display information on the monitor. Monitor resolution, Number of colors, monitor refresh rate, video card and the monitor's quality all determine the quality of the display.



7.5. Sound Card

Sound Card: To handle sound, to insert a microphone or connect a speaker this sound card is used. Many sound cards have MIDI controllers. This is also called SWSynth.



7.6. Network card

Network Interface Card: This is also referred as NIC. The computer can be connected to a network only with the use of this network interface card.



Chapter 8: I/O Connectors

8.1. Ports

Ports are physical connectors and their corresponding software that conduct signals into and out of computers. Figure 1 shows the most common ports. These are often built into the motherboard, but are also on expansion cards.



Figure 1: Common ports

There are so many connections that can be done on the computer; however, it is important to observe a rule which is a guide to how you connect your components to the System unit.

Connection Rule:

If a connection is not going in easily, then it is probably wrong. Be careful with cables do not force anything to connect.

Types of Connection on the System Unit

1. DB Connector: These are some of the most common types of connectors. They look like a D-shape and they are called D-Shape connection. They come in different sizes but they are basically the same type of connectors and they can be Male or Female DB-connectors.



Figure 2: DB Connector

Male and Female DB-connectors: Male DB-Connectors are DB-Connectors with pins sticking out while the Female DB-Connectors have holes in them.



Figure 3: The Male DB-Connector



Figure 4: The Female DB-Connector

DB-Terminology

The concept of naming a DB-connection is DB, Dash, number of pins. For example

- A. Serial Mouse is a DB-9 Connector. That is it has nine pins. 9-pin serial ports carry ting devices (mice).
- B. A parallel port used for printers is a DB-25

2. RJ Connectors

There are two types of RJ connectors. These are

- A.) RJ-45 Connection commonly used in networks
- B.) RJ-11 Connections commonly use for the telephone and modem connection. They are smaller than RJ-45 Connectors

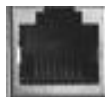


Figure 5: RJ Connector

3. USB (Universal Serial Bus)

USB means Universal Serial Bus Connections. They are for connecting different devices which include printers, keyboards, mouse, camera (digital), sound speakers etc. They are one of the most versatile types of connection.



Figure 6: USB Connector

The USB is normally represented by the symbol shown here



Symptoms and Maintenance of USB Problems

USB problems can surface when the driver for the device is not installed properly or it has become corrupted. In such a case the device you insert (e.g. printer, scanner) will not work properly or will not even be recognized by the computer.

Before you conclude that the problem is USB problem, it is important that you confirm first that the device you are attaching is working properly.

Possible Connection Problems

Usually, the USB port detects when a device (like a digital camera, keyboard etc) is attached determines automatically what host resources are needed and then makes those resources available without any user intervention. However, connection problems are possible.

The most common are:

- The USB port is disabled in the PC BIOS setup
- The PC cannot detect the USB components (USB Controller and Root Hub)
- There is a conflict with another device

4. DIN

The most common type of DIN connectors is called PS/2 connectors. They are round type of connectors that are used to connect keyboard and mouse. The PS/2 which is purple in color is used for keyboard while green one is for mouse connection. There is always an indicator to show the way the connection is to go. If keyboard is connected in the wrong position, the keyboard will not work.



Figure 7: PS/2 Connector

5. MINI AUDIO

These are used to connect Audio devices like Microphone, Speakers etc. They normally come with the sound cards.

Expansion Slots

Expansion slots consist of AGP video slots, PCI and ISA slots. PCI and ISA slots are used primarily for modems, network adapters, sound cards, additional ports (such as USB or parallel), and video cards. You will likely encounter other types of cards as well.

Removal and Installation of Expansion Cards

Removal and replacement of expansion cards is usually simple. You might want to use an anti-static wristband for this. With the computer powered off and disconnected, open the case and remove the screw (if present) holding the card in place, as shown in Figure 8.



Figure 8: Removing the screw that holds an expansion card in place.

Next, pull out the card, using only as much force as necessary. Do not bend the card. After removing the card, if you plan on reinstalling it or installing another card, you are ready. However, if you plan to leave the slot empty, you'll want to install a slot cover, as shown in Figure 9.

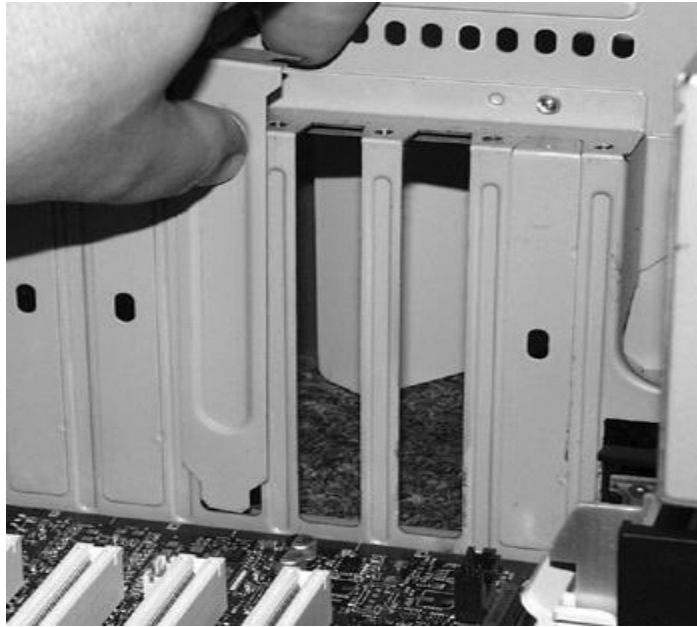


Figure 9: Installing a slot cover

To install another card, just follow the procedure in reverse.

8.1. Keyboard

Keyboard is the primary input device. It is used for entering commands and data into the system.

The primary keyboard types are as follows:

- 101-key Enhanced keyboard
- 104-key Windows keyboard
- 83-key PC and XT keyboard (obsolete)
- 84-key AT keyboard (obsolete)

Note: New systems today use the 101- or 104-key keyboard design

Enhanced 101-Key (or 102-Key) Keyboard

The layout of this universal keyboard was improved over that of the 84-key unit, with perhaps the exception of the Enter key, which reverted to a smaller size. The 101-key Enhanced keyboard was designed to conform to international regulations and specifications for keyboards. The Enhanced keyboard is available in several variations, but all are basically the same electrically.

The 101-key keyboard layout can be divided into the following four sections:

✚ Typing area

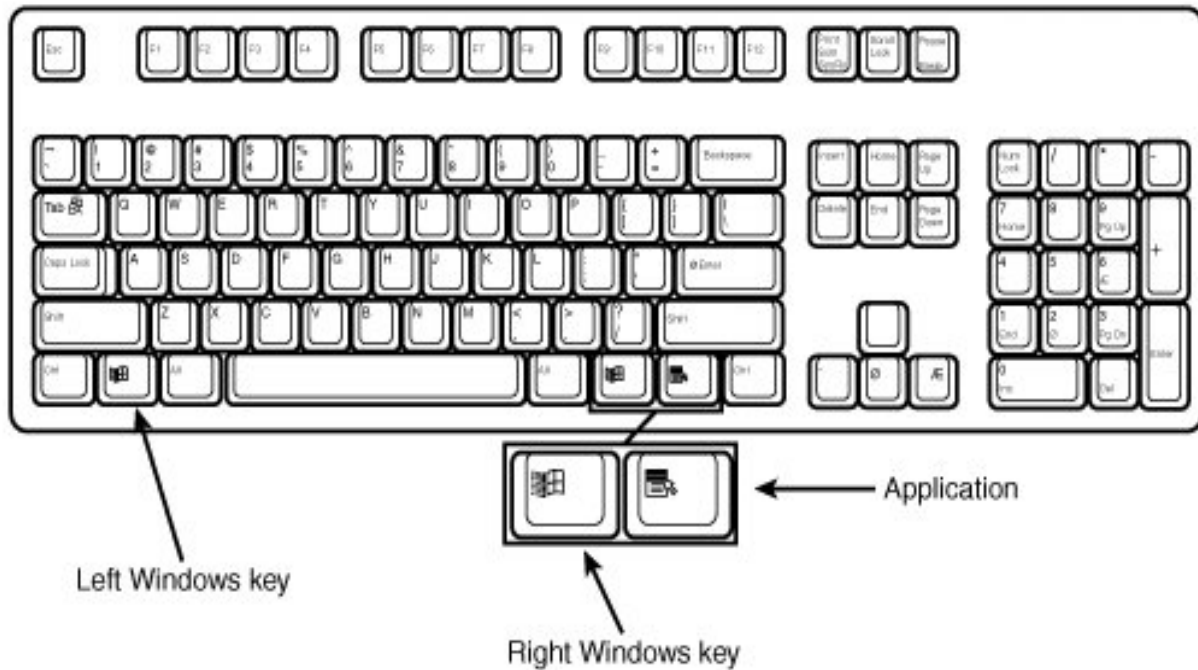
- ✚ Numeric keypad
- ✚ Cursor and screen controls
- ✚ Function keys

104-Key (Windows 9x/Me/2000/XP) Keyboard

When Microsoft released Windows 95, it also introduced the Microsoft Natural Keyboard, which implemented a revised keyboard specification that added three new Windows-specific keys to the keyboard.

The Microsoft Windows keyboard specification, which has since become standard for many desktop and laptop keyboards, outlines a set of additional keys and key combinations. The 104-key layout includes left and right Windows keys and an Application key (see Figure 1). These keys are used for operating system and application-level keyboard combinations, similar to the existing Ctrl and Alt combinations. You don't need the new keys to use Windows, but software vendors are adding specific functions to their Windows products that use the new Application key (which provides the same functionality as clicking the right mouse button). The recommended Windows keyboard layout calls for the Left and Right Windows keys (called WIN keys) to flank (be located at the sides of something) the Alt keys on each side of the spacebar, as well as an Application key on the right of the Right Windows key. Note that the exact placement of these keys is up to the keyboard designer, so variations exist from keyboard to keyboard.

Figure 1: The 104-key Windows keyboard layout



The WIN keys open the Windows Start menu, which you can then navigate with the cursor keys. The Application key simulates the right mouse button; in most applications, it brings up a context-sensitive pop-up menu. Several WIN key combinations offer preset macro commands as well. For example, you can press WIN+E to launch the Windows Explorer application. [Table1](#) shows a list of all the Windows key combinations used with the 104-key keyboard.

Table 1: Key Combinations

Key Combination	Resulting Action
WIN+R	Runs dialog box
WIN+M	Minimize All
Shift+WIN+M	Undo Minimize All
WIN+D	Minimize All or Undo Minimize All
WIN+F1	Help
WIN+E	Starts Windows Explorer
WIN+F	Find Files or Folders
Ctrl+WIN+F	Find Computer
WIN+Tab	Cycles through taskbar buttons

WIN+Break	Displays System Properties dialog box
Application key	Displays a context menu for the selected item

The Windows and Application keys are not mandatory when running Windows. In fact, preexisting standard key combinations can perform the same functions as these newer keys.

The Windows keyboard specification requires that keyboard makers increase the number of trilograms in their keyboard designs. A trilogram is a combination of three rapidly pressed keys that perform a special function, such as Ctrl+Alt+Delete. Designing a keyboard so that the switch matrix correctly registers the additional trilograms plus the additional Windows keys adds somewhat to the cost of these keyboards compared to the previous 101-key standard models.

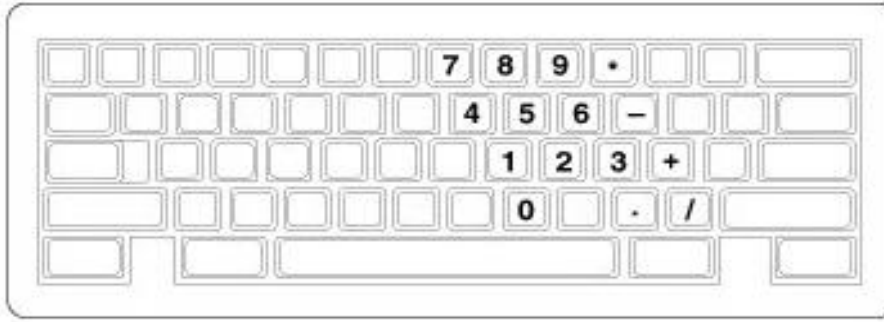
Laptop Computer Keyboards

One of the biggest influences on keyboard design in recent years has been the proliferation (rapid growth) of laptop systems. Because of size limitations, it is obviously impossible to use the standard keyboard layout for a portable computer. Manufacturers have come up with many solutions. Unfortunately, none of these solutions has become an industry standard, as the 101-key layout is. Because of the variations in design, and because a portable system keyboard is not as easily replaceable as that of a desktop system, the keyboard arrangement should be an important part of your purchasing decision.

Early laptop systems often used smaller-than-normal keys to minimize the size of the keyboard, which resulted in many complaints from users. Today, the keytops on portable systems are usually comparable in size to that of a desktop keyboard, although some systems include half-sized keytops for the function keys and other less frequently used keyboard elements.

The most obvious difference in a portable system keyboard is the sacrifice of the numeric keypad. Most systems now embed the keypad into the standard alphabetical part of the keyboard, as shown in [Figure 2](#). To switch the keys from their standard values to their keypad values, you typically must press a key combination involving a proprietary function key, often labeled **Fn**.

Figure 2: Most portable systems today embed the numeric keypad into an oddly shaped block of keys on the alphabetical part of the keyboard.



This is an extremely inconvenient solution, and many users abandon their use of the keypad entirely on portable systems. To alleviate this problem, many portable system manufacturers sell external numeric keypads that plug into the external keyboard port, a serial port, or a USB port. This is a great feature for somebody performing a lot of numeric data entry.

In addition to keypad control, the **Fn** key often is used to trigger other proprietary features in portable systems, such as toggling between an internal and external display and controlling screen brightness and sound volume.

The Keyboard Interface

A keyboard consists of a set of switches mounted in a grid or an array called the key matrix. When a switch is pressed, a processor in the keyboard identifies which key is pressed by determining which grid location in the matrix shows continuity. The keyboard processor, which also interprets how long the key is pressed, can even handle multiple keypresses at the same time. A 16-byte hardware buffer in the keyboard can handle rapid or multiple keypresses, passing each one to the system in succession.

The keyboard communicates with the main system in one of two ways:

- Through a special serial data link if a standard PS/2 keyboard connector is used
- Through the USB port

PS/2 Keyboard Interface

The PS/2 keyboard interface was introduced with the IBM PS/2 series in 1986, and is now used by all ATX and ATX-variant motherboards. PS/2 keyboards use the 6-pin mini-DIN connector (female at the PC), shown in Figure 3 and whose pinouts are described in Table 2.

Figure 3: The PS/2 keyboard connector

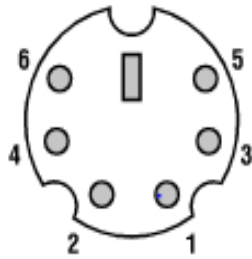


Table 2: PS/2 keyboard interface signals and pinout

Pin	Signal name	Description
1	DATA	Keyboard data
2	RESERVED	No connection
3	GROUND	Signal ground
4	VCC	+5VDC
5	CLOCK	Keyboard clock
6	RESERVED	No connection
Shield	-	Ground

USB Keyboard Interface

Some recent keyboards can be connected to a USB port. For now at least, few keyboards are USB-only. Most USB-capable keyboards also provide a standard PS/2 connector. To use a USB keyboard, the PC BIOS must support USB keyboards, and you must run an operating system, like Windows 98/98SE/Me or Windows 2000/XP, that supports USB. BIOS support is needed so that the keyboard can be used before the operating system loads to do such things as changing Setup parameters or choosing options from a boot menu. Operating system support is required for the keyboard to be accessible after the system boots.

Cleaning a Keyboard

One of the best ways to keep a keyboard in top condition is periodic cleaning. As preventive maintenance, you should vacuum the keyboard weekly, or at least monthly. When vacuuming, you should use a soft brush attachment; this will help dislodge (remove or force out from a position) the dust. Also note that many keyboards have keycaps that can come off easily. Be careful when

vacuuming; otherwise, you'll have to dig them out of the vacuum cleaner. We recommend using a small, handheld vacuum cleaner made for cleaning computers and sewing machines; these have enough suction to get the job done with little risk of removing your keytops.

You also can use canned compressed air to blow the dust and dirt out instead of using a vacuum. Before you dust a keyboard with the compressed air, turn the keyboard upside down so that the particles of dirt and dust collected inside can fall out.

Troubleshooting and Repairing Keyboards

If a keyboard stops working or behaves strangely, check to make sure the cables is connected properly. If everything appears to be correct, it's worth doing a simple swap to verify whether the problem is the keyboard or the PC. If a known-good keyboard also fails to work, the problem is most likely caused by a defective keyboard interface on the motherboard, for which the only realistic solution is to replace the motherboard. If the known-good keyboard works, replace the defective keyboard.

Mouse

A computer mouse is a small handheld device. Most mice consist of an oblong case with a roller underneath and two or three buttons on top or along the sides. When you move the mouse along a desktop, an onscreen pointer (called a cursor) moves in response. When you click (press and release) a mouse button, this motion initiates an action in your program.

Mice come in all shapes and sizes. Some have wires, and some are wireless. Some are relatively oval in shape, and others are all curvy to better fit in the palm of your hand. Some have the typical roller ball underneath, and others use an optical sensor to determine where and how much you're rolling.

Mouse Interfaces

Mice can be connected to PCs in at least one of four ways: serial (DB-9 connector), USB, and infrared or radio frequency wireless. Many mice come with adapters so that an individual mouse can be used in both PS/2 and serial ports, or both PS/2 and USB ports.

Serial: This is the original interface used in personal computers. Although new computers no longer come with serial mice, the vast majority of computers have serial ports, so a serial mouse is the perfect choice when the built-in PS/2 port fails.

PS/2: This is the standard mouse interface for the vast majority of computers with ATX and similar motherboards. These computers have dedicated PS/2 mouse ports that can be used only for PS/2 pointing devices and those adapted to PS/2.

USB: This is the standard mouse port for many new computers. However, because most computers still have PS/2 mouse ports, unless the mouse you want is available only in USB, it is a good idea to use a PS/2 mouse. This is because there are many devices that use USB ports, but only one that can use a PS/2 mouse port, so you might as well save the USB port for some other device. Most of the feature-laden mice use USB. USB devices are recognized and installed by Windows, and many come with software disks. Windows, however, has drivers for almost all commercially available mice. USB mice are also good choices for replacements when PS/2 ports fail.

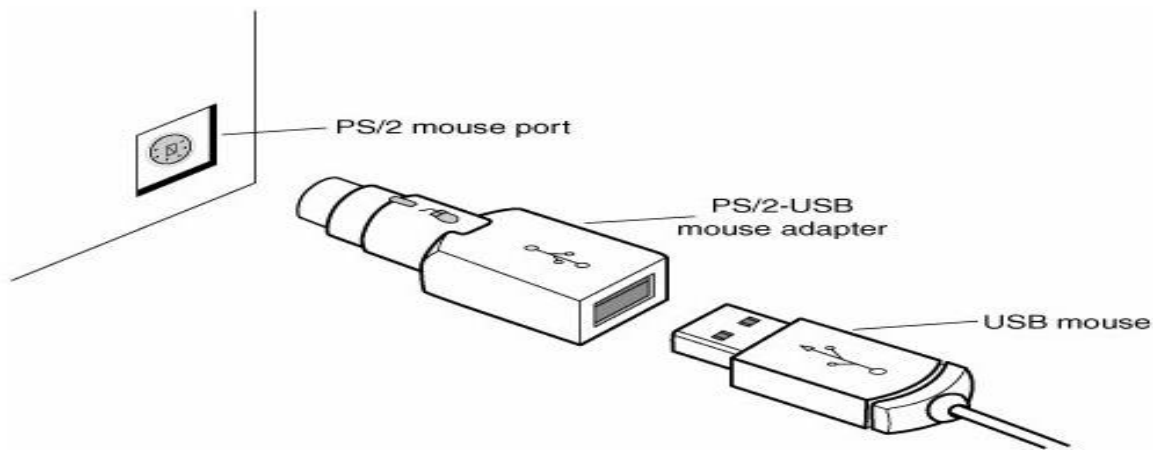
Wireless: Wireless mice use one of the other interfaces in this list. They come in two varieties, infrared and radio frequency (RF). The RF types are usually preferred because the receiver and mouse do not have to be directly in line of site of each other. The mouse runs on battery power, so plan to have spare batteries on hand or at least hold on to a basic wired mouse in case of an emergency.

Hybrid Mice

Hybrid mice are those designed to plug into two types of ports. Although a few mice are designed to plug into either the serial port or the PS/2 port, most mice are designed to plug into either the PS/2 port or the USB port.

Circuitry in a hybrid mouse automatically detects the type of port to which it is connected and configures the mouse automatically. Serial-PS/2 hybrid mice usually come with a mini-DIN connector on the end of their cable and an adapter that converts the mini-DIN to a 9-pin serial port connector, although the reverse is sometimes true on early examples of these mice. PS/2-USB mice usually come with the USB connector on the end of their cable and include a mini-DIN (PS/2) adapter, as shown in [Figure 4](#).

Figure 4: A typical USB mouse with a PS/2 adapter



Detection Types

Mice come in two detection types: mechanical (ball) and optical. Ball mice have rubber-coated metal balls that roll over a surface. This is the cheapest and most common type. It is also the most troublesome because the mechanical parts get clogged with dust, lint, and hair, and they also break. They usually need mouse pads, or at least table or desktops of a certain texture to work well.

Optical mice use a reflected red LED light to detect movement. Early optical mice sometimes required certain types of surfaces, but since then, optical mice have become easy to use and much more reliable than ball mice.

Cleaning Your Mouse

If you notice that the mouse pointer moves across the screen in a jerky fashion, it might be time to clean your mouse. For a mouse with a roller-ball, this jerkiness is caused when dirt and dust become trapped around the mouse's ball-and-roller assembly, thereby restricting its free movement.

8.3. Monitor

The images you see on a computer monitor are generated inside your computer, via a specialized video card (sometimes called a video adapter or graphics adapter). This card generates video images at a specified resolution, which is a measurement of how detailed your picture is. (The higher the resolution, the sharper the picture—and the more items you can fit onscreen.) These

video images then are displayed on your monitor. The resolution of the card must match the resolution of the monitor, or you get gibberish onscreen.

There are two different display technologies used in today's computer monitors. The older technology uses a big cathode ray tube (CRT) to display the images. The newer technology uses a thin liquid crystal display (LCD) instead of a CRT—and is quickly becoming the display of choice.

A CRT monitor, like the one in Figure 1, works and looks pretty much like a television set, but specialized to handle computer data. CRT monitors are big (depth-wise) but fairly low priced.

Figure 1: A traditional CRT monitor—big screen.



LCD monitors, in contrast, give you the same size picture in a much thinner package. Most LCD monitors, like the one in [Figure 2](#), are only a few inches thick, which makes them ideal for portable PCs and desktop systems where space is at a premium. LCD technology is more expensive than CRT technology, however, which is why you'll pay at least twice as much for a comparable LCD monitor over a CRT model—although the price is coming down.



Figure 2: LCD monitor

Size

The size of all computer monitors, like all television sets, is measured in terms of diagonal inches. That is, you measure the distance from the lower-left corner to the upper-right corner of the display.

The problem, though, is that this diagonal measurement differs between CRT monitors and LCD monitors. How is that, you ask? Just compare a 15-inch CRT monitor and a 15-inch LCD monitor, and you'll see that their screens really aren't the same size. The 15-inch LCD monitor is bigger, despite the identical specification. That's because an LCD display offers a larger viewable area; the picture goes to the very edge of the frame. In contrast, CRT monitors typically have a black fringe (the outside boundary or surface of something) around the edge of the tube, which decreases the viewable area.

So when you're comparing screen sizes, you have to do a little math in your head to recognize that a 15-inch LCD monitor offers the same size display as a 17-inch CRT monitor. Or, if you don't want to do the math, just consult the comparison information in Table 1

Table 1: Screen Size Comparison

CRT Monitor	LCD Monitor
12-inch	10-inch

14/15-inch	12-inch
17-inch	15-inch
21-inch	17-inch

Resolution

The resolution of a display is measured in pixels, which are the small dots that combine to make up the total display. The more pixels displayed, the higher the resolution—and the more information you can display onscreen.

The pixel level is specified by horizontal and vertical measurements. For example, a resolution of 640x480 displays 640 pixels horizontally (left to right) and 480 pixels vertically (top to bottom).

Note that resolution is independent of screen size; you can have a very large monitor running a low resolution, or a very small monitor running a high resolution. In general, though, if you buy a larger monitor, you'll want to make sure that it's capable of displaying a higher resolution; if not, you won't be able to see more things on your screen, just bigger things.

Video Card

There are some very good reasons for many users to upgrade their video cards. Many low-end PCs come with low-performance video cards, which just aren't up to snuff (at the required standard, of good enough quality) when it comes to displaying high-resolution graphics, fast-moving game images, and moving video from DVDs and digital movies

Resolution

Just as your monitor displays images at a specific resolution, your video card generates images at a specific resolution. Make sure the resolution of your card matches the resolution of your monitor; if you generate a higher resolution than your monitor is capable of, all you see onscreen is gibberish.

Memory

All video cards contain some amount of RAM, which is used to temporarily store graphics information while images are being assembled for display. The more memory on your video card, the faster high-resolution images (moving images, in particular) can be displayed.

If you're upgrading your video for gaming purposes, you want a card with at least 64MB of video memory. Some high-end video cards carry 128MB or even 256MB RAM, which provide extremely speedy rendering of 3D graphics.

3D Acceleration

A graphics accelerator is a special chip on the video card that assumes the task of drawing shapes onscreen. This enables faster rendering of graphics objects.

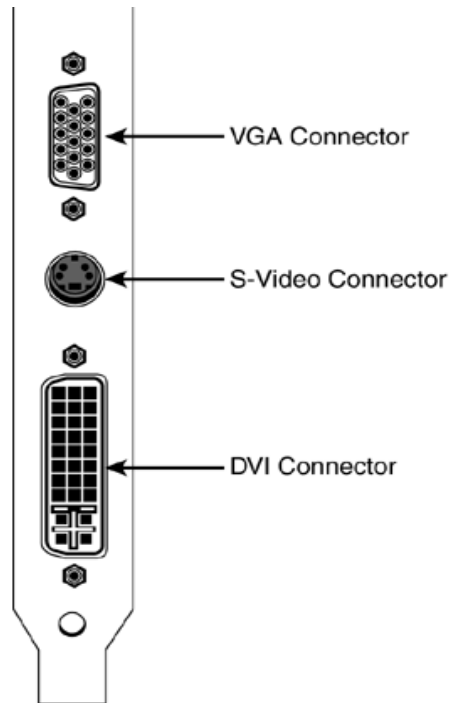
Connectors

Different types of monitors use different types of connectors. All CRT monitors and some LCD monitors use a traditional analog video (VGA) connector. Some LCD monitors use a Digital Video Interface (DVI) connector. Most video cards include a VGA connector; some include both VGA and DVI connectors.

In addition, if you want to display your computer information on a traditional television set, you'll need a card with a video output jack. This output can be a fairly low-resolution composite video jack (a single RCA connector), or the higher-resolution S-video connector.

Note: Some really high-end video cards—those capable of decoding high-definition television signals—include even higher-resolution component video connectors.

Figure 3: shows a video card with all three types of connectors: analog (VGA), digital (DVI), and video out (S-video).



Slots

You'll find video cards that insert into either PCI or AGP expansion slots. Low-priced video cards typically use PCI slots, whereas higher-priced cards use AGP slots.

Troubleshooting Monitors

Problem: No picture.

Solution

If the LED on the front of the monitor is yellow or flashing green, the monitor is in power-saving mode. Move the mouse or press Alt+Tab on the keyboard and wait for some seconds to wake up the system if the system is turned on.

If the LED on the front of the monitor is green, the monitor is in normal mode (receiving a signal), but the brightness and contrast are set incorrectly; adjust them.

If no lights are lit on the monitor, check the power and power switch. Check the surge protector or power director to ensure that power is going to the monitor. Replace the power cord with a known-working spare if necessary. Retest. Replace the monitor with a known-working spare to ensure that the monitor is the problem.

Check data cables at the monitor and video card end.

Problem: Jittery picture quality.

Solution

LCD monitors. Use display-adjustment software or onscreen menus to reduce or eliminate pixel jitter (a small irregular movement resulting from fluctuations in the voltage supply or mechanical vibrations or other sources) and pixel swim.

All monitors. Check cables for tightness at the video card and the monitor (if removable):

- ✚ Remove the extender cable and retest with the monitor plugged directly into the video card. If the extended cable is bad, replace it.
- ✚ Check the cables for damage; replace as needed.
- ✚ If problems are intermittent (stopping and starting at irregular intervals), check for interference. (Microwave ovens near monitors can cause severe picture distortion when turned on.)

CRT monitors. Check refresh-rate settings; reduce them until acceptable picture quality is achieved:

- ✚ Use onscreen picture adjustments until an acceptable picture quality is achieved.
- ✚ If problems are intermittent and can be "fixed" by waiting or gently tapping the side of the monitor, the monitor power supply is probably bad or has loose connections internally. Service or replace the monitor.

Troubleshooting Video Cards and Drivers

Problem: Display works in DOS (command-prompt mode) but not in Windows.

Solution

If you have an acceptable picture quality in MS-DOS mode (system boot) but no picture in Windows, most likely you have an incorrect or corrupted video driver installed in Windows. Boot Windows 9x/Me in Safe Mode (which uses a VGA driver), boot Windows 2000/XP in Enable VGA mode, or install the VGA driver and restart Windows. If Safe Mode or VGA Mode works, get the correct driver for the video card and reinstall.

If you have overclocked your card with a manufacturer-supplied or third-party utility, you might have set the speed too high. Restart the system in Safe Mode, and reset the card to run at its default speed. If you have adjusted the speed of AGP/PCI slots in the BIOS setup program, restart the system, start the BIOS setup program, and reset the AGP and PCI slots to run at the normal speed.

Problem: Can't replace built-in video card with add-on PCI, AGP, or PCI Express video card.

Solution

Check with the video card and system vendor for a list of acceptable replacement video cards. Try another video card with a different chipset. Check the BIOS or motherboard for jumper or configuration settings to disable built-in video. Place the add-on card in a different PCI slot. Be sure the card is fully inserted into the slot.

Problem: Can't select desired color depth and resolution combination.

Solution

Verify that the card is properly identified in Windows and that the card's memory is working properly. Use diagnostic software provided by the video card or chipset maker to test the card's memory. If the hardware is working properly, check for new drivers. Use the vendor's drivers rather than the ones provided with Windows.

Problem: Can't enable secondary monitor.

Solution

If you are using a dual-head graphics card, make sure the secondary monitor has been enabled in the manufacturer's driver. This can require you to open the Advanced settings for your driver. If you are using an AGP and a PCI video card in separate slots, check the BIOS configuration for primary VGA BIOS. Change the current setting and restart the system. Update video card drivers.

Troubleshooting Hard Drives

Operating System is Missing

One of the most obvious signs of a hard drive problem is an error message early in the boot process saying that the system can't find an OS. However, before jumping to conclusions, make sure there is no floppy in the floppy drive. It is most common for a BIOS to be set to boot first from a floppy, so if there is a non-bootable floppy in the drive, you'll probably get a message such as "Non-system disk or disk error," or "ntldr is missing." Remove the floppy and restart before taking any other measures.

If there is no floppy in the drive and you continue to get a missing OS message, it is time to check the hard drive for problems. You can try the following:

- A. Listen for hard drive activity and look at the indicator light on the case. The light should flicker and you should hear some sounds. Buzzing or clicking, however, is a possible sign of drive failure.
- B. Check the BIOS to make sure that it recognizes the hard drive. Set the BIOS to autodetect the hard drive.
- C. Check the power and data cables. Make sure they are plugged in correctly and securely. If the data cable looks damaged, try a replacement cable. Try disconnecting the power cable and using another one from the power supply.
- D. Check the hard drive jumpers. Make sure there aren't two slaves, two masters, or a mixture of slave or master and cable select.
- E. Remove the hard drive and check it on another computer. Look for the Windows folder to make sure it is intact (undamaged), and scan the drive for viruses.

Troubleshooting Memory Problems

So, how do you know when memory is a problem? First, if a problem occurs just after you installed new memory, you know that memory is a possible culprit (wrongdoing). Other signs of a memory problem include frequent [lockups](#) (when the mouse pointer won't move and the Num Lock light on the keyboard is stuck on or off), or there is suddenly no video. Of course, neither of these symptoms is a certain indication of memory problems. There can be many reasons why these things could happen, such as viruses or corrupted Windows files.

One type of error message that you'll be forced to heed is the type that appears early in the boot process and often prevents the computer from booting. Examples of these are the memory mismatch or memory parity error. They usually mean that the wrong type of memory is installed in the computer, incompatible modules are combined in the same machine, or that a module is faulty or not correctly installed.

8.4. BIOS

BIOS is a term that stands for basic input/output system, which consists of low-level software that controls the system hardware and acts as an interface between the operating system and the hardware. BIOS is essentially the link between hardware and software in a system.

When the PC was first introduced, the BIOS software containing all the device drivers for the entire system was collectively burned into one or more nonvolatile (meaning they retain their data even when the power is turned off) read-only memory (ROM) chips and placed on the motherboard. In essence, the drivers were self-contained, preloaded into memory, and accessible anytime the PC was powered on.

A PC system can be described as a series of layers: some hardware and some software that interface with each other. In the most basic sense, you can break a PC down into four primary layers, each of which can be broken down further into subsets. Figure 1 shows the four layers in a typical PC.

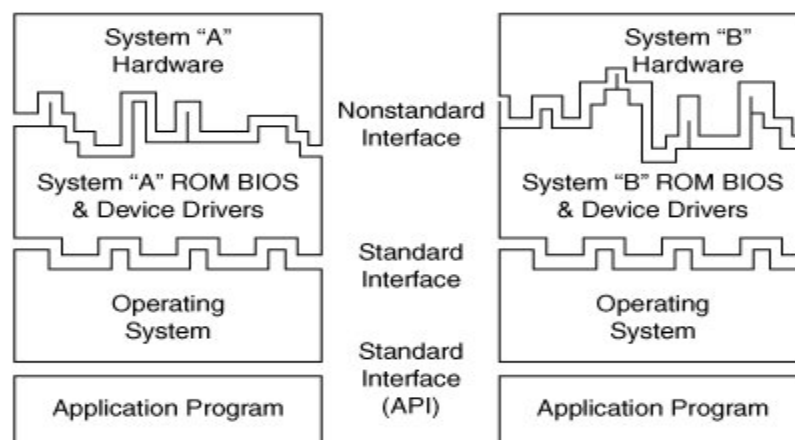


Figure 1: PC system layers

The purpose of the layered design is to enable a given operating system and applications to run on different hardware. Figure 1 shows how two different machines with different hardware can each use different sets of drivers (BIOS) to interface the unique hardware to a common operating system and applications. Thus, two machines with different processors, storage media, video display units, and so on can run the same OS and applications.

In this layered architecture, the application software programs talk to the operating system via what is called an Application Program Interface (API). The API varies according to the operating system you are using and consists of the various commands and functions the operating system can perform for an application. For example, an application can call on the operating system to load or save a file. This prevents the application itself from having to know how to read the disk, send data to a printer, or perform any other of the many functions the operating system can provide. Because the application is completely insulated from the hardware, you can essentially run the same applications on different machines; the application is designed to talk to the operating system rather than the hardware.

The operating system then interfaces with or talks to the BIOS or driver layer. The BIOS consists of all the individual driver programs that operate between the operating system and the actual hardware. As such, the operating system never talks to the hardware directly; instead, it must always go through the appropriate drivers. This provides a consistent way to talk to the hardware. It is usually the responsibility of the hardware manufacturer to provide drivers for its hardware. Because the drivers must act between both the hardware and the operating system, the drivers typically are operating system specific. Thus, the hardware manufacturer must offer different drivers so that its hardware works under DOS, Windows 9x, Windows 2000, Windows XP, Linux, and so on. Because many operating systems use the same internal interfaces, some drivers can work under multiple operating systems. For example, a driver that works under Windows XP will also often work under Windows 2000 and NT (and vice versa).

Because the BIOS layer looks the same to the operating system no matter what hardware is above it (or underneath, depending on your point of view), the same operating system can run on a variety of systems. For example, you can run Windows XP on two systems with different processors, hard disks, video adapters, and so on, yet Windows XP will look and feel pretty much the same on both

of them. This is because the drivers provide the same basic functions no matter which specific hardware is used.

As you can see from Figure 1, the application and operating systems layers can be identical from system to system, but the hardware can differ radically. Because the BIOS consists of software drivers that act to interface the hardware to the software, the BIOS layer adapts to the unique hardware on one end but looks consistently the same to the operating system at the other end.

The hardware layer is where most differences lie between various systems. It is up to the BIOS to mask the differences between unique hardware so that the given operating system (and subsequently the application) can be run.

The BIOS itself is software running in memory that consists of all the various drivers that interface the hardware to the operating system. The BIOS is unique compared to normal software in that it doesn't all load from disk; some of it is preloaded into memory chips (read-only memory, or ROM) installed in the system or on adapter cards.

The BIOS in a PC comes from three possible sources:

-  Motherboard ROM
-  Adapter card ROM (such as that found on a video card)
-  Loaded into RAM from disk (device drivers)

The motherboard ROM BIOS is most often associated with hardware rather than software. This is because the BIOS on the motherboard is contained in a ROM chip on the board, which contains the initial software drivers needed to get the system running.

Motherboard BIOS

All motherboards must have a special chip containing software called the ROM BIOS. This ROM chip contains the startup programs and drivers used to get the system running and act as the interface to the basic hardware in the system. When you turn on a system, the power on self test (POST) in the BIOS also tests the major components in the system. Additionally, you can run a setup program to store system configuration data in the CMOS memory, which is powered by a battery on the motherboard.

The BIOS is a collection of programs embedded in one or more chips, depending on the design of your computer. That collection of programs is the first thing loaded when you start your computer, even before the operating system. Simply put, the BIOS in most PCs has four main functions:

POST (power on self-test). The POST tests your computer's processor, memory, chipset, video adapter, disk controllers, disk drives, keyboard and other crucial components.

Setup. The system configuration and setup program is usually a menu-driven program activated by pressing a special key during the POST, and it enables you to configure the motherboard and chipset settings along with the date and time, passwords, disk drives, and other basic system settings. You also can control the power-management settings and boot-drive sequence from the BIOS Setup, and on some systems, you can also configure CPU timing and clock-multiplier settings.

Bootstrap loader. A routine that reads the first physical sector of various disk drives looking for a valid master boot record (MBR). The MBR program code then continues the boot process by reading the first physical sector of the bootable volume, which is the start of the volume boot record (VBR). The VBR then loads the first operating system startup file, which is usually IO.SYS (DOS/Windows 9x/Me) or NTLDR (Windows NT/2000/XP), upon which the operating system is then in control and continues the boot process.

BIOS (basic input/output system). This refers to the collection of actual drivers used to act as a basic interface between the operating system and your hardware when the system is booted and running. When running DOS or Windows in safe mode, you are running almost solely on ROM-based BIOS drivers because none are loaded from disk.

ROM Hardware

Read-only memory (ROM) is a type of memory that can permanently or semi permanently hold data. It is called read-only because it is either impossible or difficult to write to. ROM is also often called nonvolatile memory because any data stored in ROM remains even if the power is turned off.

Running or accessing the CMOS Setup Program

If you want to run the BIOS Setup program, you usually have to reboot and press a particular key or key combination during the POST. Normally, the correct key to press is displayed onscreen momentarily during the POST. The major vendors have standardized the following keystrokes to enter the BIOS Setup in recent systems:

- ✚ AMI BIOS. Press Delete during POST.
- ✚ Phoenix BIOS. Press F2 during POST.
- ✚ Award BIOS. Press Delete or Ctrl+Alt+Esc during POST.
- ✚ Microid Research (MR) BIOS. Press Esc during POST.

If your system does not respond to one of these common keystroke settings, you might have to contact the manufacturer or read the system documentation to find the correct keystrokes.

The BIOS information's are stored in a CMOS chip which is kept under constant voltage supply by the computer backup battery. This way the information's contained in the CMOS are always alive even if the computer power is turned off.

The listing below is showing all the information's stored into the BIOS;

- ✚ Time and Date
- ✚ Number of Floppy Disk Drives
- ✚ Number of Hard Disk Drives
- ✚ Number of CD-ROM Drives
- ✚ Boot sequence (Enable the user to decide what disk will be checked first when booting)
- ✚ Cache Memory information's (size, type, timing, etc)
- ✚ Main Memory Information's (size, type, timing, etc)
- ✚ Setting of PCI and ISA slots
- ✚ AGP Port Settings (aperture size, etc)
- ✚ Virus Protection Warning
- ✚ Password Protection (enable the user to set his password)

When you power on your computer, the motherboard's basic input/output system (BIOS) first detects the computer's internal hardware configuration and then performs the power-on self-test (POST). The POST executes several diagnostic tests that check the basic functionality of the power

supply, motherboard, CPU, I/O Bus Controller, RAM and video components. If the computer passes the POST, it will usually return a single short beep or two short beeps, depending on the BIOS manufacturer. If the POST detects any problems, it will halt the boot process and return an error message either visually on the screen and/or audibly through a series of beeps.

Unfortunately, there is no official standard for beep codes. It depends on the BIOS manufacturer. you should always consult your motherboard's documentation, the manufacturer's Web site and the BIOS manufacturer's Web site for a definitive beep code list for your particular BIOS.

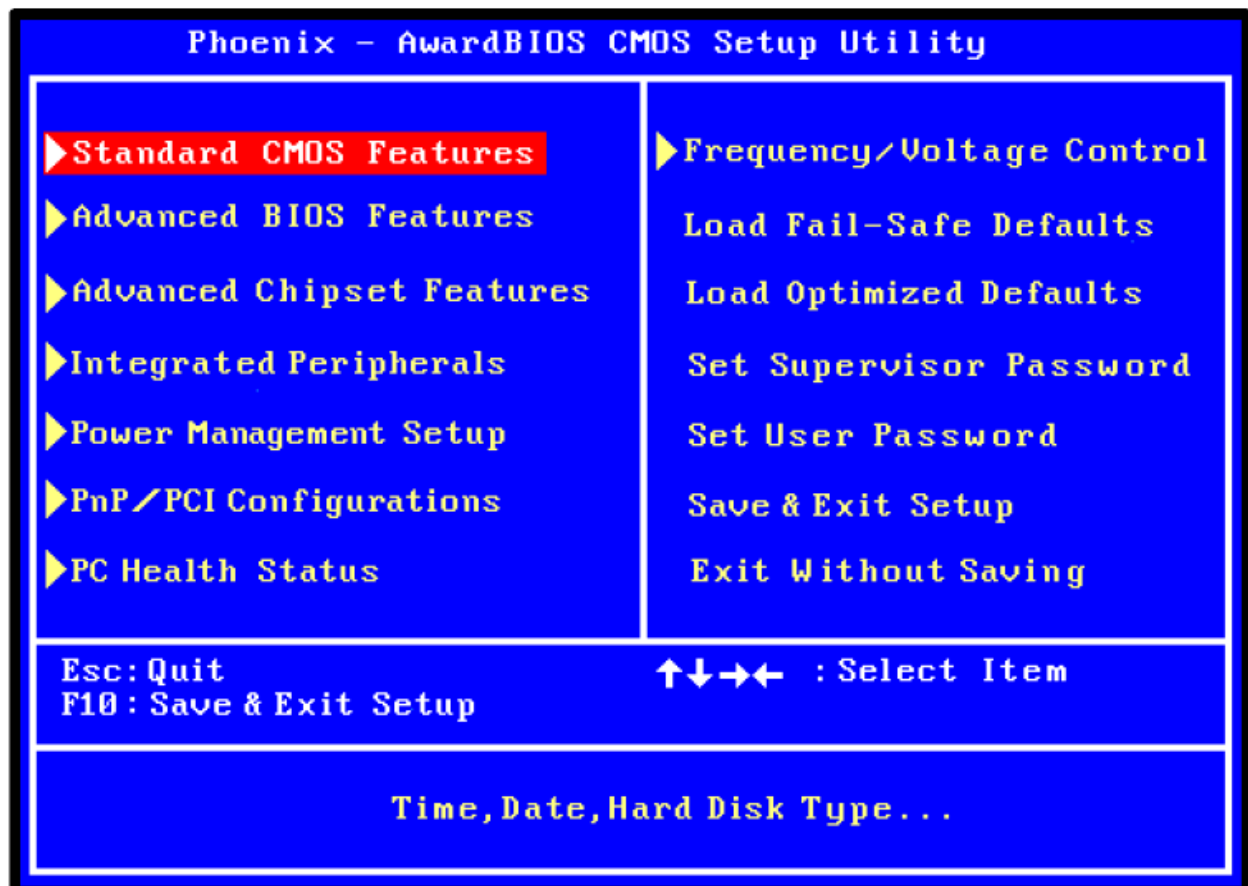


Figure 2: Award BIOS CMOS Setup

When a PC system is first powered on, the system runs a POST. If errors are encountered during the POST, you usually see a text error message displayed onscreen. Errors that occur very early in the POST might happen before the video card is initialized. These types of errors can't be displayed, so the system uses two other alternatives for communicating the error message. One is beeping the system beeps the speaker in a specific pattern that indicates which error has occurred.

Whenever you start up your computer, it automatically performs a series of tests that checks the primary components in your system, such as the CPU, ROM, motherboard support circuitry, memory, and major peripherals such as the expansion chassis. These tests are brief and are designed to catch hard (not intermittent) errors. The POST procedures are not very thorough compared with available disk-based diagnostics. The POST process provides error or warning messages whenever it encounters a faulty component.

Although the diagnostics performed by the system POST are not very thorough, they are the first line of defense, especially when it comes to detecting severe motherboard problems. If the POST encounters a problem severe enough to keep the system from operating properly, it halts the system boot process and generates an error message that often identifies the cause of the problem. These POST-detected problems are sometimes called fatal errors because they prevent the system from booting.

How Errors Are Displayed

The POST tests normally provide three types of output messages: audio codes, onscreen text messages, and hexadecimal numeric codes that are sent to an I/O port address.

POST errors can be displayed in the following three ways:

- ✚ Beep codes. Heard through the speaker attached to the motherboard.
- ✚ POST checkpoint codes. Hexadecimal checkpoint codes sent to an I/O port address. A special card plugged into either an ISA or a PCI card slot is required to view these codes.
- ✚ Onscreen messages. Error messages displayed onscreen after the video adapter is initialized.

BIOS POST Beep Codes

Beep codes are used for fatal errors only, which are errors that occur so early in the process that the video card and other devices are not yet functional. Because no display is available, these codes take the form of a series of beeps that identify the faulty component. When your computer is functioning normally, you should hear one short beep when the system starts up at the completion of the POST, although some systems (such as Compaq's) beep twice at the end of a normal POST.

If a problem is detected, a different number of beeps sounds, sometimes in a combination of short and long tones.

BIOS POST Checkpoint Codes

POST checkpoint codes are hexadecimal numeric codes written by POST routines to I/O port. These are often simply called POST codes. These POST codes can be read by only a special adapter card plugged into one of the system slots. Several companies make these cards available to technicians.

POST checkpoint codes can be used to track the system's progress through the boot process from power-on right up to the point at which the bootstrap loader runs (when the operating system load begins). When you plug a POST code reader card into a slot, during the POST you will see two-digit hexadecimal numbers flash on the card's display. If the system stops unexpectedly or hangs, you can identify the test that was in progress during the hang from the two-digit code. This step usually helps to identify the malfunctioning component.

BIOS POST Onscreen Messages

Onscreen messages are brief messages that attempt to indicate a specific failure. These messages can be displayed only after the point at which the video adapter card and display have been initialized.

These different types of error messages are BIOS dependent and vary among BIOS manufacturers, and even in some cases among different BIOSs from the same manufacturer. You should consult your motherboard or ROM BIOS manufacturer for the codes specific to your board and BIOS.

8.5.2. Error messages and solutions

Blue Screen of Death—this error screen is the most infamous scene that can appear on your Windows computer screen. This occurs when the operating system encounters a “stop error,” which is caused by damaged or faulty hardware. In some cases, the computer will automatically restart as a form of defense.

Response: A user can boot the computer in “safe mode” and then perform a clean boot process on the OS. Reinstallations will have to be made and drivers should be updated to avoid these errors in the future. Unfortunately, a BSOD error will cause data loss in any situation.

The application *insert name here* is not open anymore—this error message is seen on Mac computers, even in the newer models. It means that even if an app is open, it is currently unresponsive and cannot be used. Some users attribute this problem to the RAM on the device.

Response: A user can stop the app, exit out of it, and try to launch it again. This may work in some circumstances, but if the issue persists, it may be a sign of something more serious within your computer’s operation.

Boot Device Not Found—this message appears for Windows users because a boot device is required for the computer to start properly and the machine isn’t recognizing any device. The causes of this failure can be many things including the hard drive’s boot sector is corrupt, the BIOS was rearranged in a configuration change, or the hard drive itself may be damaged.

Response: You can start by checking the BIOS settings and ensure the boot order is correct. If this is not the issue, you may need to replace your hard drive and install a new version of your Windows operating system. In this instance, you will have no access to the files on the hard drive that was removed and will need data recovery.

Unable to install application—this message appears when you attempt to install or download a new file or program. It means that there is not enough free space in the internal storage to accommodate the new data.

Response: A user needs to free space by deleting files and folders that are no longer needed. In some cases, the OS may be missing updates and the applications are not installing. Optimizing your hard drive and updating the computer will help to solve this issue. Regular updating and a backup strategy will ensure that you do not experience unnecessary data loss.

SMART Hard Disk Error 301—this error indicates that the hard disk or solid-state drive has already experienced a failure, or will soon. This error message will appear when you turn on the

device and can cause serious damage if not treated immediately. It could be the result of a broken controller chip, failed installation of an application, a power surge, or malware.

Response: Sometimes, a user can change the BIOS sequence or attempt a reboot, but if the drive has a physical error, then it is best not to run the computer to avoid further damage.

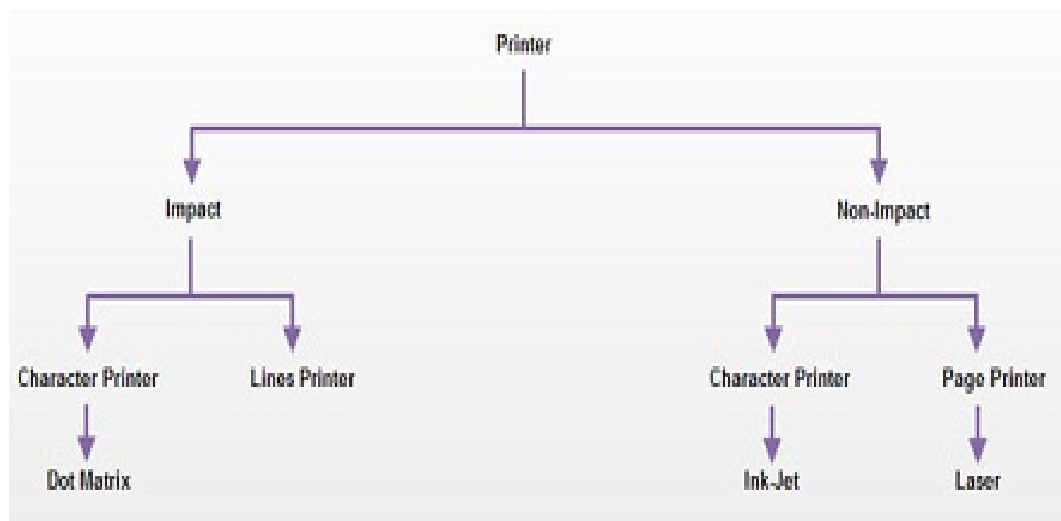
8.6. Printers

Printers are the most commonly used and popular output devices. They provide information in a permanent readable form. They produce printed output of results, programs and data. Prints text or image on paper called hard copy. The printing speed of a printer is measured in pages per minute (ppm). Printer resolution is often measured in dots per inch (dpi).

8.6.1. Printer Types and Printer Technology

Printers are Output devices used to prepare permanent Output devices on paper. Printers can be divided into two main categories:

Impact Printers: In this hammers or pins strike against a ribbon and paper to print the text. This mechanism is known as electro-mechanical mechanism. They are of two types.



8.1. Impact Printers

An impact printer makes contact with the paper. It usually forms the print image by pressing an inked ribbon against the paper using a hammer or pins. Following are some examples of impact printers.

8.1.1. Dot-Matrix Printers

The dot-matrix printer uses print heads containing from 9 to 24 pins. These pins produce patterns of dots on the paper to form the individual characters. The 24-pin dot-matrix printer produces more dots than a 9 pin dot-matrix printer, which results in much better quality and clearer characters. The general rule is: the more pins, the clearer the letters on the paper. The pins strike the ribbon individually as the print mechanism moves across the entire print line in both directions, i-e, from left to right, then right to left, and so on. The user can produce a color output with a dot-matrix printer (the user will change the black ribbon with a ribbon that has color stripes). Dot-matrix printers are inexpensive and typically print at speeds of 100-600 characters per second.

8.1.2. Daisy-wheel printers

In order to get the quality of type found on typewriters, a daisy-wheel impact printer can be used. It is called daisy-wheel printer because the print mechanism looks like a daisy; at the end of each “Petal” is a fully formed character which produces solid-line print. A hammer strikes a “petal” containing a character against the ribbon, and the character prints on the paper. Its speed is slow typically, 25-55 characters per second.

8.1.3. Line printers

In business where enormous amount of material are printed, the character-at-a-time printers are too slow; therefore, these users need line-at-a-time printers. Line printers, or line-at-a-time printers, use special mechanism that can print a whole line at once; they can typically print the range of 1,200 to 6,000 lines per minute. Drum, chain, and band printers are line-at-a-time printers.

8.1.4. Drum printer

A drum printer consists of a solid, cylindrical drum that has raised characters in bands on its surface. The number of print positions across the drum equals the number available on the page. This number typically ranges from 80-132 print positions. The drum rotates at a rapid speed. For each possible print position there is a print hammer located behind the paper. These hammers strike the paper, along the ink ribbon, against the proper character on the drum as it passes. One revolution of the drum is required to print each line. This means that all characters on the line are not printed at exactly the same time, but the time required to print the entire line is fast enough to call them line printers. Typical speeds of drum printers are in the range of 300 to 2000 lines per minute.

8.1.5. Chain printers

A chain printer uses a chain of print characters wrapped around two pulleys. Like the drum printer, there is one hammer for each print position. Circuitry inside the printer detects when the correct character appears at the desired print location on the page. The hammer then strikes the page, pressing the paper against a ribbon and the character located at the desired print position. An impression of the character is left on the page. The chain keeps rotating until all the required print positions on the line have filled. Then the page moves up to print the next line. Speeds of chain printers range from 400 to 2500 characters per minute.

8.1.6. Band printers

A band printer operates similar to chain printer except it uses a band instead of a chain and has fewer hammers. Band printer has a steel band divided into five sections of 48 characters each. The hammers on a band printer are mounted on a cartridge that moves across the paper to the appropriate positions. Characters are rotated into place and struck by the hammers. Font styles can easily be changed by replacing a band or chain.

8.2. Non-impact printers

Non-impact printers do not use a striking device to produce characters on the paper; and because these printers do not hammer against the paper they are much quieter. Following are some non-impacted printers.

8.2.1. Ink-jet printers

Ink-jet printers work in the same fashion as dot-matrix printers in the form images or characters with little dots. However, the dots are formed by tiny droplets of ink. Ink-jet printers form characters on paper by spraying ink from tiny nozzles through an electrical field that arranges the charged ink particles into characters at the rate of approximately 250 characters per second. The ink is absorbed into the paper and dries instantly. Various colors of ink can also be used. One or more nozzles in the print head emit a steady stream of ink drops. Droplets of ink are electrically charged after leaving the nozzle. The droplets are then guided to the paper by electrically charged deflecting plates [one plate has positive charge (upper plate) and the other has negative charge (lower plate)]. A nozzle for black ink may be all that's needed to print text, but full-color printing is also possible with the addition of needed to print text, but full-color printing is also possible with the addition three extra nozzles for the cyan, magenta, and yellow primary colors. If a droplet isn't needed for the character or image being formed, it is recycled back to its input nozzle. Several

manufacturers produce color ink-jet printer. Some of these printers come with all their color inks in a cartridge; if you want to replace one color, you must replace all the colors. Other color ink-jet printers allow you to replace ink individually. These printers are a better choice if user uses one color more than other colors. These printers produce less noise and print in better quality with greater speed.

8.2.2. Laser printers

A laser printer works like a photocopy machine. Laser printers produce images on paper by directing a laser beam at a mirror which bounces the beam onto a drum. The drum has a special coating on it to which toner (an ink powder) sticks. Using patterns of small dots, a laser beam conveys information from the computer to a positively charged drum to become neutralized. From all those areas of drum which become neutralized, the toner detaches. As the paper rolls by the drum, the toner is transferred to the paper printing the letters or other graphics on the paper. A hot roller bonds the toner to the paper.

Laser printers use buffers that store an entire page at a time. When a whole page is loaded, it will be printed. The speed of laser printers is high and they print quietly without producing much noise. Many home-use laser printers can print eight pages per minute, but faster and print approximately 21,000 lines per minute, or 437 pages per minute if each page contains 48 lines. When high speed laser printers were introduced, they were expensive. Developments in the last few years have provided relatively low-cost laser printers for use in small businesses.

8.6.2. Printer Field Replaceable Units

field replacement unit or “fru” means any component or sub-assembly of an item or unit of hardware which can reasonably be replaced at a customer location. field replacement unit or "fru" shall mean a field replaceable unit, or those parts, options, and/or components that can be used to affect a fix on a product.